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The Ride-Hailing and Pricing Strategy Decision With the Application of Autonomous Vehicles

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ABSTRACT

The application of autonomous vehicles (AVs) offers a promising avenue to address the profitability challenges of ride-hailing platforms. However, in a mixed operational model combining human drivers (HDs) and AVs, the high operating costs of AVs raise concerns about their feasibility for large-scale commercial deployment and the design of optimal pricing strategies. To address these issues, this study develops a cooperative game-theoretic model involving three parties: a ride-hailing platform, an AV manufacturer, and a jointly established company. We first examine changes in platform profitability before and after AV application to identify the conditions under which adopting AVs becomes viable. We then compare the payoffs of all market participants under different pricing strategies to determine the optimal scheme that facilitates multiparty cooperation. The results indicate that introducing AVs can strengthen user-platform stickiness and reduce wage costs for HDs. However, the platform is incentivized to adopt AVs only when the operating cost coefficient is sufficiently low. Moreover, once this cost falls below a certain threshold, a premium pricing strategy yields higher platform profitability and enables a win–win–win outcome among the platform, joint company, and manufacturer. This study advances the literature on economic decision-making for AV deployment in the ride-hailing sector and offers theoretical and practical guidance for pricing strategies and partnership design in the era of intelligent mobility.

1 | Introduction

As the mobile internet and smart devices become increasingly ubiquitous, ride-hailing platforms have experienced a dramatic surge in popularity worldwide, revolutionizing the transportation landscape and redefining the future of urban mobility (Choi et al. 2020; Jiang et al. 2021). However, despite their massive success in terms of user base and service coverage, profitability remains a significant challenge. The primary issue is the substantial expenditure on human drivers (HDs), which encompasses commissions, incentive fees, and an array of insurance and benefits. These expenses consume a substantial portion of platforms' revenue, making profitability a daunting goal. Using autonomous vehicles (AVs) is one way to address these issues; thus, many ride-hailing platforms view AVs as a promising solution for enhancing their profitability (Siddiq and Taylor 2022). For example, companies such as Uber and Lyft have already invested billions of dollars in the autonomous driving sector.[1]

In recent years, significant advancements in autonomous driving technology have brought AVs into the public eye (Baron et al. 2022; Mo, Chen, et al. 2022; Sharma and Rana 2024).

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The integration of AVs into ride-hailing fleets holds the potential to deliver profound societal advantages. This technology is capable of alleviating urban traffic congestion through sophisticated route optimization, improving traffic flow efficiency, decreasing the incidence of accidents, enriching passenger experiences, and reducing carbon footprints (Stogios et al. 2019; Yu et al. 2021). Some important examples of AV operation are as follows: As of April 2023, Didi Chuxing's AVs had operated safely for 1020 consecutive days.[2] By April 19, 2024, Baidu's Apollo Go had provided 6 million rides to the public.[3] According to China's People's Daily, Beijing plans to support the use of AVs for ride-hailing services.[4] Additionally, at least 18 cities in the United States are either operating or piloting AVs, with 23 states having passed laws allowing the strategy or deployment of AVs.[5] According to McKinsey, fully AVs will be able to operate "anytime, anywhere" by 2030.[6] However, ensuring the safe operation of AVs hinges on their ability to precisely perceive and interpret their surroundings in real time and to rapidly transmit data. This requirement necessitates substantial investments in technology development and large-scale operation (Freund et al. 2022). Therefore, the strategic decision on whether to integrate AVs into ride-hailing services looms large for these platforms' plans.

Additionally, introducing AVs into ride-hailing services presents a highly challenging engineering endeavor, considering the requirements of technology development and manufacturing (Badue et al. 2021). Currently, ride-hailing platforms are increasingly seeking strategic partnerships with AV manufacturers. This collaborative approach aims to leverage the technological expertise and production capabilities of AV manufacturers, thereby enhancing platforms' offerings and service quality. For example, in 2023, Uber and Waymo forged a strategic partnership,[7] integrating Waymo's autonomous driving technology into Uber and establishing a hybrid network of HDs and AVs. Uber had previously collaborated with Volvo as early as 2016 on AV development.[8] In China, Didi Chuxing's autonomous driving unit and Guangzhou Automobile Group's Aion launched a project for the mass production of new energy AVs, forming a joint company.[9] Leveraging Aion's large-scale production capabilities, the joint company aims to introduce the first commercially available AV by 2025 and integrate it into Didi's ride-hailing network, facilitating a large-scale mixed dispatch system of HDs and AVs.

In a mixed dispatch model that includes both HDs and AVs, the pricing strategies for ride-hailing platforms become significantly more complex. Ride-hailing platforms must factor in passengers' reception of AVs, along with sophisticated matching algorithms, production and operating expenses, and the financial impacts of technological research and development (R&D) and software updates by joint companies. These elements play a critical role in shaping an effective pricing model in a complex, mixed-fleet environment. Additionally, platforms must delicately balance the profitability of manufacturers and joint companies to foster enduring strategic alliances, all while maximizing their financial returns. Crafting a pricing strategy that seamlessly integrates both HDs and AVs is a sophisticated and pivotal challenge. This strategy must not only appeal to customers but also guarantee operational efficiency and financial viability for the platforms, manufacturers, and joint companies. According to reports, during the trial phase of AV operations, many platforms set fares that are higher than regular ride-hailing services if discounts are not applied, such as Baidu's AV company.[10] In contrast, Didi Chuxing applies standardized taxi pricing for its commercial operations in Huadu, Guangzhou.[11] AVs eliminate the need for driver commissions, thereby increasing the profitability of ride-hailing platforms (Siddiq and Taylor 2022). However, the application of AVs entails substantial operating costs, increasing the financial burden on platforms (Castro and Frazelle 2024). Thus, whether AVs should adopt the same pricing standards as HDs remains an unanswered question.

Against this backdrop and motivated by the discussions above, we develop a cooperative game model involving a ride-hailing platform, an AV manufacturer, and a joint company established by both parties. The ride-hailing platform first strategically decides whether to introduce AVs into its fleet. If not, only HDs provide services. If the decision is to introduce AVs, the platform collaborates with the AV manufacturer to establish a joint company that is responsible for the R&D and operational tasks of AVs, while the manufacturer handles production. Therefore, the joint company determines the number of AVs to be deployed in the market, and the manufacturer sets the original equipment manufacturer (OEM) price. Subsequently, the platform operates a hybrid fleet of HDs and AVs. Based on the ride price for HDs, the platform can choose between two pricing strategies for AVs: same pricing or differentiated pricing (including discounted or premium pricing). Our research addresses the following four questions:

1. How do fluctuations in market dynamics, such as network externalities among passengers, commission rates, and investment distributions, influence equilibrium outcomes?
2. How does the application of AVs affect the ride-hailing market (including ride prices, HD supply, and passenger demand)?
3. Under what conditions will the platform introduce AVs?
4. Which pricing strategy is more beneficial for the platform's profitability and for achieving a win–win situation for all parties involved?

Through this research, we make several key contributions. We find that the operating costs borne by the joint company play a critical role in the strategic introduction of AVs and sensitivity analysis. Contrary to intuition, when operating costs increase, the ride-hailing platform and AV manufacturer do not raise prices to obtain marginal profits; instead, they lower prices to mitigate the reduction rate of AV deployment. Consistent with reality, high operating costs are detrimental to AV adoption. Additionally, regardless of the pricing strategy employed, the application of AVs leads to some HDs exiting the market, thus reducing wage expenses for the platform. However, only when the operating cost coefficient of AVs is relatively low does the platform have an incentive to introduce them. Post-introduction, when the operating cost coefficient falls below a certain threshold, a premium pricing strategy is more profitable for the platform while ensuring mutual benefits for the platform, joint company, and manufacturer. If the operating cost coefficient exceeds this threshold, a discounted pricing strategy yields

better profitability for the platform, although it creates conflicts of interest with the joint company and manufacturer. Therefore, the same pricing strategy is more favorable for long-term collaboration between the platform and manufacturer. From a social welfare perspective, only when the operating cost coefficient is extremely low does a premium pricing strategy enhance overall social welfare. In the extended model, we consider additional factors such as the same-side network externalities for HDs, AV production costs, heterogeneous investment proportions between the platform and manufacturer, and passengers' sentiment toward AVs. Overall, these extensions validate the stability of the basic model.

Related research focuses primarily on the application of AVs in ride-hailing, pricing strategies for ride-hailing platforms, and cooperative and competitive dynamics within ride-hailing platforms. Specifically, this paper bridges the gap between commercial practice and theoretical research. Our main novel contributions are as follows:

1. While many studies explore the application of AVs in ride-hailing (Al-Kanj et al. 2020; Castro and Frazelle 2024; Mo, Chen, et al. 2022; P. Wang et al. 2025a; X. Wang et al. 2021; Xie et al. 2025; Yao et al. 2020; Yi and Smart 2021), few examine whether platforms should introduce AVs from a game-theoretic perspective. We address this issue based on economic and social benefits, clarifying the corresponding conditions and contributing to a deeper understanding of AV adoption by ride-hailing platforms.
2. In research on pricing strategies for ride-hailing platforms, most studies focus on ride-pricing strategies (Bai and Tang 2022; Benjaafar et al. 2022; H. Feng et al. 2025; Y. Feng et al. 2023; Hu et al. 2024; Y. Li et al. 2021; Nourinejad and Ramezani 2020; Taylor 2018; J.-H. Zhang et al. 2025). This study goes beyond this focus and examines pricing strategies for AVs, benchmarking them against the ride price set by human-driven services. Doing so not only provides theoretical support for enhancing platform profitability in the future large-scale commercial deployment of AVs but also offers a new perspective for fostering sustainable cooperation between platforms and manufacturers.
3. Research on cooperative models for ride-hailing platforms does not consider the introduction of AVs (T. Dong et al. 2021; S. Feng et al. 2022; Fielbaum et al. 2025; Ke et al. 2021; B. Li, Zhao, et al. 2023; Lin et al. 2021; F. Yang et al. 2023; L. Yang et al. 2025; Zhou et al. 2022). We innovatively model the latest business model of mixed dispatching of HDs and AVs as a game, propose a cooperative model of establishing a joint company between a ride-hailing platform and an AV manufacturer, and explore its impact on the ride-hailing market.

The rest of this paper is structured as follows. In Section 2, we conduct a comprehensive review and summary of the literature related to this study. Section 3 provides a detailed depiction of the notations, entities, and sequence of events in the model. In Section 4, we present the equilibrium outcomes for each strategy and analyze the impact of changes in the market environment on these equilibrium outcomes. On this basis, Section 5 discusses the strategic choices available to the ride-hailing platform in the AV business. Section 6 proposes four extensions to augment and refine the basic model. Finally, Section 7 summarizes the paper. The proofs of all equilibrium outcomes, propositions, and corollaries are detailed in the Supporting Information.

2 | Literature Review

We have identified three main streams of literature related to our study: (1) the application of AVs in ride-hailing; (2) pricing strategies of ride-hailing platforms; and (3) cooperative models in ride-hailing platforms. The following sections provide a detailed review of the relevant literature.

2.1 | Application of AVs in Ride-Hailing

A growing body of literature investigates the application of AVs in ride-hailing. Empirical studies have shown that AVs can effectively address the increasing demands of urban transportation (Y. Dong et al. 2025; Wu and Mvondo 2025; Yao et al. 2025). Many researchers focus on the operation and management of AVs within ride-hailing platforms, particularly dynamic planning and algorithms. For example, Al-Kanj et al. (2020) investigate a ride-hailing system comprising AVs and devise a high-quality operational scheduling strategy based on approximate dynamic planning and determined ride prices through an adaptive learning approach. Yu et al. (2021) address the operational planning challenges involving scheduling, relocating, and charging decisions for AVs. They formulate the problem as a Markovian decision-making process and propose an asynchronous learning approach to resolve it. Regarding mixed fleets of AVs and HDs on ride-hailing platforms, Ao et al. (2024) propose a parameterized theoretical model to characterize the decision-making behaviors of HDs, passengers, and the platform. Their results indicate that the mixed fleet reduces passenger waiting times, lowers travel costs, and increases platform profits. P. Wang et al. (2025) use a mixed-integer linear programming model to analyze intercity ride-hailing operations with AVs; findings suggest that an excessively large fleet may decrease the platform's total revenue. Additionally, some scholars have examined the economic impacts of AVs on the ride-hailing market. Siddiq and Taylor (2022) capture decisions about ride prices, wages, and AV fleet size in a game-theoretic model and explore whether the introduction of AVs by ride-hailing platforms can gain a competitive advantage. Mo, Chen, et al. (2022) investigate the complex landscape of ride-hailing services, considering congestion externalities and passenger heterogeneity in perceived utility. Their market equilibrium analysis examines how fleet size (both HDs and AVs) and prices influence demand rates and passenger waiting times. In the context of mixed operations involving autonomous and human-driven vehicles, Xie et al. (2025) develop a multi-agent reinforcement learning approach to implement differentiated pricing based on service type, time, and location, thereby maximizing benefits for both the supply and demand sides. However, Castro and Frazelle (2024) construct and solve a game model for the ride-hailing platform, incorporating both AVs and HDs. Their findings indicate that higher costs associated with AVs may pose challenges for the platform. Benjaafar et al. (2025) examine the impact of AVs on the strategic behavior and welfare of HDs and characterize the platform's optimal decisions using

a fluid model. In addition to exploring competition between AVs and HDs, Lian and van Ryzin (2024) also examine competition among independent AV operators in their analysis of the ride-hailing market.

Previous studies have primarily examined the operation of AVs in ride-hailing platforms from an algorithmic perspective, while game-theoretic research has mainly focused on competition between AVs and HDs. In contrast, our study constructs a game-theoretic model to examine whether the platform should introduce AVs and identifies the conditions under which this is optimal. Furthermore, we analyze operational decisions, including ride pricing under uniform and differentiated pricing strategies, OEM prices for AVs, and AV fleet size. We also investigate how market dynamics, such as passenger network externalities, influence these decisions.

2.2 | Pricing Strategies of Ride-Hailing Platforms

The pricing strategies of ride-hailing platforms constitute a key focus of this paper. Dynamic pricing within ride-hailing platforms has emerged as a prominent research area (Besbes et al. 2021; Bimpikis et al. 2019; Castillo et al. 2025; Shi et al. 2024; Sun and Liu 2023). For instance, H. Wang et al. (2025) explore the design of dynamic pricing strategies to achieve optimal outcomes in markets where ride- and non-pooling services coexist. Bai, So, et al. (2019) propose an analysis model with endogenous supply and demand in their study on the pricing strategies of ride-hailing platforms, incorporating steady-state waiting time performance based on a queueing model into the customer utility function. Subsequently, Tripathy et al. (2023) and Bai, Heese, et al. (2023) examine driver collusion in their dynamic pricing studies of ride-hailing platforms, each proposing two mitigation strategies to reduce the harm of such behavior on platforms and passengers. Moreover, pricing strategies play a crucial role in the business model choices of ride-hailing platforms. J.-H. Zhang et al. (2025) develop a two-stage framework to investigate how B2C ride-hailing platforms coordinate services between regular and luxury vehicles. Their results demonstrate that jointly optimizing fleet scheduling and pricing strategies can enhance the system's long-term welfare. H. Feng et al. (2025) study the business model selection of ride-hailing platforms, comparing the closed model—where the platform uses its vehicles to serve passengers—with the open model, which allows private vehicles to provide services. They reveal that the open model increases service supply during peak periods and reduces price surges, benefiting passengers. Additionally, some literature focuses on the pricing strategies of competing ride-hailing platforms. C. Zhang et al. (2022) develop a model comparing multiple wage schemes, such as fixed commission rates, dynamic commission rates, and fixed wages, across two competing platforms. Their findings highlight that each of the three wage schemes may have opposing effects on the platforms, consumers, and HDs. Cai et al. (2023) employ a two-period Stackelberg model to characterize the competition between two heterogeneous ride-hailing platforms, where the leader platform operates both NEVs and fuel vehicles, and find that the leader's pricing can directly and indirectly affect supply. Similarly, Zhao et al. (2023) model the competition between two heterogeneous ride-hailing platforms within the context of Peak Carbon. Besides interplatform competition, ride-hailing platforms represent significant competitors for traditional taxis. Yu et al. (2020) study the impact of government regulatory policies on the welfare of various stakeholders in the competition between the ride-hailing platform and taxis. They provide pricing outcomes of the platforms in a 2-cycle dynamic game, demonstrating that well-designed regulatory policies can achieve equilibria satisfying multiple competing objectives. Additionally, Zhong et al. (2022) investigate the competitive pricing strategies and government regulatory strategies of the ride-hailing platform and the taxi industry under unregulated and regulated pricing scenarios.

Most existing studies have focused solely on pricing strategies for HDs providing ride-hailing services. In contrast, our research not only examines the pricing decisions of HDs but also analyzes the pricing strategies following the application of AVs. Specifically, we investigate how AV fares should be set relative to those charged by HDs and whether a unified pricing standard should be maintained. Furthermore, we investigate the impacts of different pricing strategies for AVs on HDs and overall social welfare. Additionally, we develop a multi-agent cooperative game model involving AV operations, which has not been addressed in prior literature.

2.3 | Cooperative Models in Ride-Hailing Platforms

Our paper is closely connected to research on cooperative models for ride-hailing platforms. Various studies have explored this topic from diverse perspectives. Lin et al. (2021) investigate whether the ride-hailing platform should collaborate with the car rental company to offer rental services for individuals without cars, considering consumer and driver surplus. They find that the platform might be more inclined to cooperate with the car rental company. T. Dong et al. (2021) concentrate on the cooperation model between the ride-hailing platform and HDs, including freelancers and contract HDs, and examine the optimal contract design under demand uncertainty, freelancer price sensitivity, and variations in HDs' risk attitudes. Similarly, Fielbaum et al. (2025) examine the cooperation between ride-hailing platforms and drivers, finding that idle wages can serve as a complementary mechanism to coordinate supply and demand. Additionally, Ke et al. (2021) extend the scope of ride-hailing platforms to multimodal transportation systems, including public transportation. They analyze the complementary and substitution relationships between the ride-hailing platform and public transportation, highlighting the pivotal role of fleet size on the platform in shaping this relationship. Similarly, S. Feng et al. (2022) also focus on intermodal systems and model real-time order scheduling to coordinate ride-hailing and public transportation services as a large-scale sequential decision-making problem. Furthermore, the integration of ride-hailing platforms represents an innovative business model (Cao et al. 2025). Zhou et al. (2022) devise a mathematical modeling framework to depict multiple competitive platforms and compare system performance under two market scenarios: with or without a third-party integrator under Nash equilibrium and social welfare. The existing literature has primarily explored platform competition rather than cooperation or integration (Bernstein et al. 2021; Mo, Yu, et al. 2020). Addressing the importance of NEVs in ride-hailing platform strategies, B. Li, Zhao, et al. (2023) discuss four potential

TABLE 1 | Comparison between this study and the existing literature.

Literature	AV	Pricing	Cooperation	Strategy selection
Ao et al. (2024)	√	√		
Siddiq and Taylor (2022)	√	√		√
Mo, Chen, et al. (2022)	√	√		
Xie et al. (2025)	√	√		
Castro and Frazelle (2024)	√	√		√
Tripathy et al. (2023)		√		√
H. Feng et al. (2025)		√		√
Lin et al. (2021)		√	√	√
Fielbaum et al. (2025)		√	√	
B. Li, Zhao, et al. (2023)		√	√	√
This study	√	√	√	√

strategies for integration and cooperation among the ride-hailing platform, NEV manufacturer, and charging service provider, utilizing NEVs as a connecting point. In the field of shared mobility, Chen et al. (2025) investigate the channel selection problem between battery manufacturers and shared mobility platforms. This study provides valuable insights for exploring cooperative models of ride-hailing platforms in our research.

Previous studies on cooperation in ride-hailing platforms have primarily focused on partnerships between platforms and drivers, or between platforms and rental companies or aggregators in terms of business models. Few studies have examined the cooperation between platforms and manufacturers in promoting AVs. Unlike existing literature, this paper abstracts the cooperation model between the platform and manufacturer—establishing a joint company for AV services—into a game-theoretic framework to explore the impact of AVs on ride-hailing. In this model, both the platform and manufacturer invest a portion of the profits related to the AV business into the joint company. Finally, this study provides reliable pricing strategy recommendations for the platform to achieve long-term win–win cooperation among all three parties.

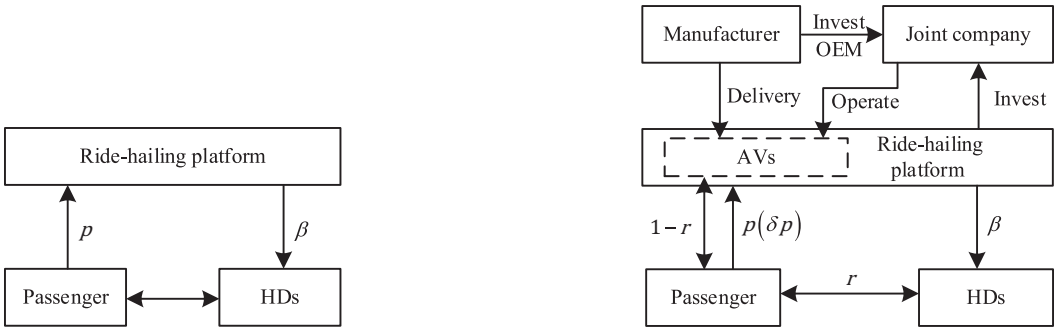
In summary, while prior research has provided insights into AV operations, pricing, and cooperation within ride-hailing platforms, our study offers a novel perspective by examining the conditions for AV adoption, pricing strategies, and multiparty cooperation. This provides valuable guidance for platform managers considering the large-scale commercial deployment of AVs and related operational decisions. Table 1 summarizes the distinctions between this study and existing literature.

3 | Model Setup

Considering the current market environment and future development trends, ride-hailing platforms need to make a strategic decision on whether to introduce AVs to enhance the passenger experience and operational efficiency. In practice, in some cities, ride-hailing platforms, in addition to providing services with human-driven vehicles, are currently deploying or planning to deploy AVs to meet passenger demand. For example, Uber operates a mixed fleet that includes AVs in Phoenix.[12] Additionally, Didi Chuxing users can experience AV services in demonstration areas in Guangzhou and Shanghai, and Didi Chuxing also plans to integrate mass-produced AVs into its ride-hailing network by 2025.[2] If the ride-hailing platform decides to introduce AVs, it must also determine the appropriate pricing strategy. In this study, we use the superscripts $\{n, s, d\}$ to denote scenarios where AVs are not introduced (strategy n), introduced with the same pricing strategy (strategy s), and introduced with a differentiated pricing strategy (strategy d), respectively. When the ride-hailing platform chooses to introduce AVs, the decision-makers include the ride-hailing platform (hereafter referred to as the platform), the AV manufacturer (hereafter referred to as the manufacturer), and the joint company funded by both parties. This paper builds the model on a unit-time basis, which is common in previous research (Benjaafar et al. 2022; T. Dong et al. 2021; Lin et al. 2021). For example, participants—HDs, the platform, manufacturer, and joint company—make decisions aimed at daily profits. Figure 1 illustrates the relationships between the involved entities, and Table 2 lists notations and definitions used in the model. The subsequent section describes in detail the decision-making behavior of each entity when the platform introduces AVs.

1. HDs

The platform operates a mixed fleet consisting of human-driven vehicles and AVs managed by a joint company to provide ride-hailing services to passengers. Assume that the number of HDs registered on the platform is denoted as K. HDs' decisions to provide services are influenced by various factors, including income, operating costs, and reservation wages. Initially, we assume that an HD's income follows the common commission-based model. The average ride price set by the platform for each transaction is represented as p, and the commission rate received by HDs is denoted as β, where $0 < \beta < 1$. When AVs are not introduced, the service times that an HD can provide per unit time is t; thus the HD's income per unit time is $t\beta p$. When



a. Not introducing AVs

b. Introducing AVs

FIGURE 1 | Relationships among entities. (a) Not introducing AVs. (b) Introducing AVs.

TABLE 2 | Definitions of notations.

Notation	Definition	Notation	Definition
Decision variables		α	The network externalities of passenger supply to HDs in the platform
p	The ride price paid to the platform by the passenger	δ	The differentiated ratio in ride prices for AVs and HDs
q	The quantity of AVs launched by the joint company	ϕ	The ratio of the platform or manufacturer's investment in the joint company
w	The OEM price paid by the joint company to the manufacturer for AVs	Endogenous variables	
Exogenous variables		U_d	Utility for HDs
β	The commission rate offered by the platform to HDs	U_c	Utility for passengers
f	Operating costs paid by HDs in the platform per unit time	k	The supply number of HDs
t	The service times of HDs per unit time in scenarios where AVs are not introduced	D	Passenger demand
ε	The reservation wage of HDs	π_p	The profit of the platform
τ	Operating cost coefficient for AVs	π_c	The profit of the joint company
λ	The number of potential passengers	π_m	The profit of the manufacturer
r	The probability that a passenger is matched to an HD when introducing AVs	DS	Driver surplus
v	Passengers' initial perceived value of the platform	CS	Passenger surplus

AVs are introduced, whether passengers can be matched with an AV depends on factors such as platform control, geographical limitations, and passenger preferences. For example, in Uber's operations, whether a passenger uses an AV is usually not directly decided by the passenger but assigned by the platform based on vehicle availability and suitability. For example, in Uber's partnership with Waymo in Phoenix, when passengers request specific service types (e.g., UberX, Uber Green, Uber Comfort, or Uber Comfort Electric), the platform might match passengers with AVs if AVs are available in that area.[13] Similarly, Didi Chuxing plans to integrate mass-produced AVs into its ride-hailing network by 2025, adopting a mixed dispatch model.[2] Following previous studies (Balakrishnan et al. 2014; Huang et al. 2024; Niu, Xie, and Chen 2024), we assume that the probability that the platform matches passengers with HDs is r, and that the probability that passengers are matched with AVs is $1 - r$. Passengers who prefer AVs can increase their chances of being matched by updating their preferences on the ride-hailing platform's app.[13] In this context, r not only represents the probability of matching passengers with HDs but also indirectly

reflects passengers' preference for AVs, indicating the ride-hailing market's acceptance of AVs. Therefore, compared with the scenario without AVs, an HD's income per unit time decreases to $rt\beta p$. Assume that the operating cost for HDs per unit time is f. Thus, the utility for HDs after introducing AVs is $U_d = rt\beta p - f - \varepsilon$, where ε represents the reservation wage of an HD, which varies among HDs. In reality, HDs estimate their potential incomes before joining the platform and will choose to be active only if these incomes exceed their reservation wage. Assume that the reservation wage ε is uniformly distributed in the range $[0, 1]$, which is a common assumption in previous research (Tripathy et al. 2023; Wang and Zhang 2022; Yu et al. 2020). Without loss of generality, we normalize the number of registered HDs to 1, i.e., $K = 1$. Consequently, based on $U_d > 0$, the number of active HDs can be derived as $k = \int_0^{\widetilde{\varepsilon}} 1 d\varepsilon = rt\beta p - f$, where $\widetilde{\varepsilon} = rt\beta p - f$. Furthermore, the driver surplus is $DS = \int_0^{\widetilde{\varepsilon}} rt\beta p - f - \varepsilon d\varepsilon = \frac{1}{2}(rt\beta p - f)^2$.

2. Passengers

When passengers decide whether to choose the platform as a travel mode, their decision is influenced by their initial perceived value, the ride price, and the vehicle supply on the platform. Let the number of potential passengers in the market be denoted as λ. Additionally, assume that passengers will always consider both choices (i.e., HDs and AVs) when AVs are introduced. Therefore, when the platform adopts the same pricing strategy, a passenger's utility is $U_c^s = v - p + \alpha(k + q)$, where v represents the passenger's initial perceived value to the platform (Bai, Heese, et al. 2023; Tripathy et al. 2023), and $v \sim U[0, 1]$; q is the quantity of AVs deployed by the joint company in the ride-hailing market; and $\alpha \in (0, 1)$ denotes the passenger's network externality coefficient to the platform's vehicle supply, meaning that more vehicles on the platform can lead to shorter wait times and higher utility for passengers. A passenger will choose the platform only when his or her utility is positive. From $U_c^s > 0$, the passenger demand under the same pricing strategy can be obtained as $D^s = \lambda \int_{\widetilde{v}}^1 1 dv = \lambda(1 - p + \alpha(k + q))$, where $\widetilde{v} = p - \alpha(k + q)$. The passenger surplus is $CS^s = \lambda \int_{\widetilde{v}}^1 v - p + \alpha(k+q) dv = \frac{\lambda}{2}(1 - p + \alpha(k+q))^2$.

When the platform adopts a differentiated pricing strategy, a passenger's utility is $U_c^d = v - (rp + (1 - r)\delta p) + \alpha(k + q)$, with δ denoting the differentiated ratio in the ride price charged for AVs and HDs. This assumption is also used in studies on differentiated pricing (Ji et al. 2023; J. Li, Ou, and Cao 2023). The pricing difference between AVs and HDs considers multiple factors, including operating costs, market demand, and the passenger experience, reflecting the platform's comprehensive balance in optimizing the service and user experience. If $0 < \delta < 1$, it indicates a discounted pricing strategy, where the ride price for AVs is lower than that for HDs. If $\delta > 1$, it indicates a premium pricing strategy, where the ride price for AVs is higher. Therefore, $rp + (1 - r)\delta p$ represents the expected ride price based on the probability of matching with an HD or AV. From $U_c^d > 0$, the passenger demand under the differentiated pricing strategy is $D^d = \lambda \int_{\widehat{v}}^1 1 dv = \lambda(1 - (rp + (1 - r)\delta p) + \alpha(k + q))$, where $\widehat{v} = rp + (1 - r)\delta p - \alpha(k + q)$. The passenger surplus is $CS^d = \lambda \int_{\widehat{v}}^1 v - (rp + (1-r)\delta p) + \alpha(k+q) dv = \frac{\lambda}{2}(1 - (rp + (1-r)\delta p) + \alpha(k+q))^2$.

3. The manufacturer

When the platform chooses to introduce AVs, it collaborates with the manufacturer to establish a joint company focused on the initial research and subsequent operation of AVs. The joint company commissions the manufacturer to mass-produce AVs, which are then handed over to the joint company and integrated into the ride-hailing platform for operation. For example, Didi Chuxing and Guangzhou Automobile Group's Aion jointly invested in a company to develop AVs, with Aion providing vehicle platform support.[9] Assume that the OEM price per AV is denoted as w. Without loss of generality, production costs paid by the manufacturer are ignored in this discussion. The case in which production costs are considered is addressed in the extended model. Therefore, the manufacturer's revenue in the OEM process is wq. In accordance with related studies on equity cooperation between enterprises (He et al. 2024; Liu and Zhou 2024; Ren et al. 2021), the manufacturer will invest a portion of its revenue in the joint company to establish cooperation with the platform, ensuring the joint company's smooth operation. This investment not only creates a new profit model for both the ride-hailing platform and the manufacturer but also strengthens their partnership and promotes the development and expansion of the AV business. Assuming that the investment ratio of the manufacturer is denoted as ϕ, where $0 < \phi < 1$, this investment can be interpreted as a discount on the OEM price that the manufacturer offers to the joint company. Thus, the manufacturer's profit function can be expressed as $\pi_m = (1 - \phi)wq$.

4. The platform

The platform profits by providing ride-hailing services through a mixed fleet of HDs and AVs. Additionally, to enhance the passenger experience, the platform aims to meet all passenger ride demands (Bernstein et al. 2021; Lin et al. 2021). In the basic model, we consider the mechanism of cooperation between the platform and the manufacturer in establishing the joint company as follows: both parties invest an equal ratio of revenue from the AV business in the joint company. Specifically, the platform contributes a proportion ϕ of the profits generated from AV-based ride-hailing services to the joint company. This investment can be regarded as a profit-sharing mechanism between the joint company and the platform. In the extended model, we discuss the scenario in which the investment ratios of the manufacturer and the platform differ. Therefore, when the platform adopts the same pricing strategy, the profit function can be expressed as $\pi_p^s = (1 - \beta)rpD^s + (1 - \phi)(1 - r)pD^s$. When the platform employs a differentiated pricing strategy, the profit function takes the form of $\pi_p^d = (1 - \beta)rpD^d + (1 - \phi)(1 - r)\delta pD^d$. In these profit functions, $(1 - \beta)rpD^s$ or $(1 - \beta)rpD^d$ represents the platform's remaining revenue after paying the HDs' commission. Additionally, the platform invests a ratio ϕ of revenue from AV services into the joint company, so the profit from AV services is $(1 - \phi)(1 - r)pD^s$ or $(1 - \phi)(1 - r)\delta pD^d$.

5. The joint company

When the platform aims to integrate AVs into its ride-hailing service, the joint company will go beyond commissioning the manufacturer to produce AVs. The joint company will also

have to work toward the large-scale commercialization of AVs. In the initial stages, the joint company is responsible for the R&D of AVs, which includes sensors, chips, operating systems, and network technologies. Additionally, during the operation of AVs, the joint company needs to analyze and store real-time data to optimize the regional distribution and ensure accurate route planning, thereby improving operational efficiency. As Lyft has described, AVs are "asset-intensive".[1] Therefore, we assume that the operating costs for the joint company are τq^2, where τ is the operating cost coefficient. This quadratic growth in investment costs has been widely adopted in related research (Fan et al. 2023; Niu, Xie, and Chen 2024; Subramanian et al. 2007) and has been validated in empirical studies (Hartman et al. 1997). Consequently, the quantity of AVs deployed in ride-hailing will be determined by the joint company. The primary source of the joint company's profit is the combined investments made by the manufacturer and ride-hailing platform. The manufacturer's investment level is ϕwq, while the platform's investment level is $\phi(1-r)pD^s$ or $\phi(1-r)\delta pD^d$. After the OEM price of AVs (wq) and operating costs (τq^2) are deducted, the joint company's profit can be calculated. Thus, when the platform adopts the same pricing strategy, the joint company's profit function is $\pi_c^s = \phi(1-r)pD^s + \phi wq - wq - \tau q^2$. When the platform employs a differentiated pricing strategy, the joint company's profit function is $\pi_c^d = \phi(1-r)\delta pD^d + \phi wq - wq - \tau q^2$. Notably, in practice, the platform and manufacturer typically inject an initial capital investment when establishing the joint company. Without loss of generality, we normalize this initial investment to zero in the model, as this exogenous capital injection does not affect the joint company's optimal decision-making within the context of the game-theoretic framework.

6. Sequence of events

The sequence of events within the entire game model is illustrated in Figure 2. First, the platform determines whether to introduce AVs. If the platform doing so, it then directly decides the ride price p. However, if the decision is to introduce AVs, the platform chooses to adopt a same pricing strategy or a differentiated pricing strategy. Subsequently, the manufacturer determines the OEM price w, followed by the joint company deciding the quantity of AVs q to be launched. Finally, the platform decides the ride price p.

4 | Model Analysis

Our analysis focuses on the platform's strategies regarding the application of AVs and pricing. Sections 4.1–4.3 derive the equilibrium outcomes for each decision-maker under the three strategies. We then examine how market environment parameters influence these equilibrium outcomes.

4.1 | Strategy n

In this subsection, the platform refrains from introducing AVs, aligning with the prevalent business state observed in the current ride-hailing market. In this state, only HDs provide ride-hailing services to passengers, with no platform-manufacturer collaboration. Therefore, within the model setup detailed in Section 3, the HD supply, demand, and profit functions for the platform under strategy n can be derived by excluding parameters related to AVs from each function. Specifically, these functions are denoted as $k^n = t\beta p - f$, $D^n = \lambda(1-p+\alpha k^n)$, and $\pi_p^n = (1-\beta)pD^n$. Under strategy n, the platform directly determines the ride price p to maximize profits. By solving, all optimal outcomes are outlined in Table 3.

To ensure the validity of the aforementioned optimal outcomes, it is essential that $\frac{2f}{t(1+\alpha f)} < \beta < \frac{1}{t\alpha}$. This requirement indicates that the platform must maintain control of the commission rate within a specific range to guarantee profitability. For example, many existing platforms observe commission rates for HDs ranging from 70% to 85%.[14] Platforms such as Uber and Ola have set commission rates of approximately 80% (Zhong et al. 2022); whereas, since the implementation of the "Sunshine Action" initiative in China, the commission rates disclosed by ride-hailing platforms have ranged from 70% to 82%.[15] Furthermore, the derivation of this constraint leads to $t\beta - f > 0$, indicating that an HD's revenue from providing ride-hailing services exceeds the operating costs that he or she incurs.

Proposition 1. *The impacts of an increase in each parameter on the optimal outcomes are shown in Table 4.*

Proposition 1 analyzes how changes in the market environment under strategy n affect the optimal outcomes. Higher network

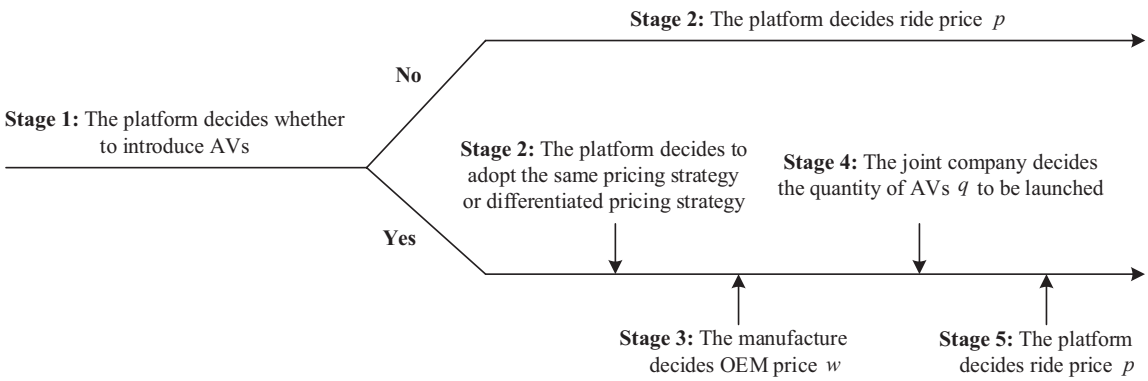


FIGURE 2 | The sequence of events.

TABLE 3 | The optimal outcomes under strategy n.

Strategy n							
p^{n*}	$\frac{1-\alpha f}{2(1-t\alpha\beta)}$	D^{n*}	$\frac{\lambda(1-\alpha f)}{2}$	k^{n*}	$\frac{t\beta-f(2-t\alpha\beta)}{2(1-t\alpha\beta)}$	π_p^{n*}	$\frac{\lambda(1-\beta)(1-\alpha f)^2}{4(1-t\alpha\beta)}$

externalities and commission rates increase platform profitability, whereas high vehicle operating costs for HDs can erode it. Specifically, higher network externalities prompt the platform to raise ride prices to attract more HDs. However, because passengers focus more on ride prices than on passenger-side network externalities, fare increases have a stronger negative impact on their utility. This, in turn, leads to a decreased demand from passengers. Conversely, when higher commission rates raise fares, the resulting negative price effect is offset by the positive utility from increased HD supply, keeping passenger demand stable. In both market scenarios, the platform can earn higher marginal profits through increased ride prices. However, higher vehicle operating costs reduce HD supply and diminish passenger experience. In response, the platform may lower ride prices to compensate passengers, but its profits may still decline.

4.2 | Strategy s

After choosing to implement a strategy for AV introduction, the platform must also determine a pricing strategy. This subsection discusses a pricing strategy in which the ride price for AVs aligns with that for HDs. Under strategy s, the profit functions for the platform, joint company, and manufacturer are outlined as follows:

$$\pi_p^s = (1-\beta)rpD^s + (1-\phi)(1-r)pD^s \tag{1}$$

$$\pi_c^s = \phi(1-r)pD^s + \phi wq - wq - \tau q^2 \tag{2}$$

$$\pi_m^s = (1-\phi)wq \tag{3}$$

We employ the backward induction method to address the profit maximization problem, and the equilibrium outcomes are presented in Table 5. For a detailed description of the solution process, please refer to the Supporting Information.

Under strategy s, beyond the criteria set for the parameters in strategy n, an additional requirement is imposed on the operating cost coefficient, denoted as $\tau > \overline{\tau} = \frac{\alpha^2\lambda\phi(1-r)}{4(1-rt\alpha\beta)}$. Although this threshold marks the platform's entry point for introducing AVs, implying a significant financial commitment to support their operating costs, it also ensures the coexistence of AVs and HDs on the ride-hailing platform. This is essential for maintaining market stability, ensuring profitability for all parties (Biswas et al. 2023; Niu, Zhang, and Zhang 2024; X. Zhang et al. 2024), and facilitating the deployment of AVs into the market. In the sensitivity analysis for strategy s, we focus specifically on outcomes that differ from those observed under strategy n. The abbreviations for parameter thresholds in the subsequent analysis are summarized in Table A1.

Proposition 2. *The impacts of an increase in the network externality coefficient α on the equilibrium outcomes are illustrated in Table 6.*

TABLE 4 | Sensitivity analysis of the optimal outcomes in the strategy n.

	p^{n*}	k^{n*}	D^{n*}	π_p^{n*}
α	↑	↑	↓	↑
β	↑	↑	—	↑
f	↓	↓	↓	↓

Similar to Proposition 1, when AVs are introduced under the same pricing strategy, stronger passenger-side network externalities affect the market in several key ways. The platform responds by increasing ride prices to ensure adequate HD supply, and the joint company strategically deploys more AVs, while the manufacturer takes the opportunity to raise the OEM price of AVs. However, the effects of network externalities on passenger demand, platform profit, and joint company profit are more complex, with the AV fleet size acting as a leverage role. Specifically, when the τ is relatively low, expanding the AV fleet shifts the leverage toward greater positive utility for passengers, thereby leading to increased passenger demand as network externalities strengthen. By contrast, in instances where τ is high, a large increase in the quantity of AVs imposes substantial costs on the joint company. Under this financial constraint, the joint company increases the fleet only marginally. This marginal increase fails to offset the negative utility from higher fares, reducing passenger demand. Consequently, when passengers are less sensitive to vehicle supply and τ remains high, both the platform and joint company incur profit losses. This is because, at this point in the profit structure, the marginal benefit from higher ride prices outweighs the market loss from reduced passengers.

Proposition 3. *The impacts of an increase in the matching probability r between passengers and HDs on the equilibrium outcomes are shown in Table 7.*

Proposition 3 examines how the equilibrium outcomes respond to changes in the matching probability r between passengers and HDs. First, a higher r incentivizes more HDs to participate in the ride-hailing market. Conversely, the joint company reduces AV deployment to relieve cost pressures, which in turn lowers the profits of both the joint company and the manufacturer as r increases. Notably, a higher r produces opposite effects on HD supply and AV quantity, which strongly shape ride price changes. Specifically, when the operating cost coefficient is low and the initial quantity of AVs is substantial, a rise in r triggers a sharp decline in the quantity of AVs. In response, the platform lowers ride prices to retain passenger demand despite the reduced supply. When the operating cost coefficient is high and the initial AV quantity is limited, a higher r causes only a small drop in AV numbers relative to the total vehicle supply. However, HD supply increases more significantly. Consequently, the platform

TABLE 5 | The equilibrium outcomes under the strategy s.

Strategy s			
w^{s*}	$\frac{\alpha\lambda\phi(1-r)(1-\alpha f)}{4(1-\phi)(1-rt\alpha\beta)}$	D^{s*}	$\frac{\lambda(1-\alpha f)\left(8(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}{4\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}$
q^{s*}	$\frac{\alpha\lambda\phi(1-r)(1-\alpha f)}{2\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}$	π_p^{s*}	$\frac{\lambda(1-\alpha f)^2(1-\phi-r(\beta-\phi))\left(8(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)^2}{16(1-rt\alpha\beta)\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)^2}$
p^{s*}	$\frac{(1-\alpha f)\left(8(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}{4(1-rt\alpha\beta)\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}$	π_c^{s*}	$\frac{\lambda\phi(1-r)(1-\alpha f)^2\left(16(1-rt\alpha\beta)\tau-3\alpha^2\lambda\phi(1-r)\right)}{16(1-rt\alpha\beta)\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}$
k^{s*}	$\frac{rt\beta(1-\alpha f)\left(8(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}{4(1-rt\alpha\beta)\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}-f$	π_m^{s*}	$\frac{(\alpha\lambda\phi(1-r)(1-\alpha f))^2}{8(1-rt\alpha\beta)\left(4(1-rt\alpha\beta)\tau-\alpha^2\lambda\phi(1-r)\right)}$

TABLE 6 | The impacts of increased α on equilibrium outcomes.

	q^{s*}	w^{s*}	p^{s*}	k^{s*}
α	↑	↑	↑	↑
	D^{s*}	π_p^{s*}	π_c^{s*}	π_m^{s*}
α	If $\overline{\tau}<\tau\le\tau_1$, ↑; otherwise, ↓	When $\alpha<\frac{2}{rt\beta}-\frac{1}{f}$, if $\tau>\tau_2$, ↓; otherwise, ↑	When $\alpha<\frac{2}{rt\beta}-\frac{1}{f}$, if $\tau>\tau_3$, ↓; otherwise, ↑	↑

TABLE 7 | The impacts of increased r on the equilibrium outcomes.

	q^{s*}	w^{s*}	p^{s*}	k^{s*}
r	↓	↓	If $\tau\le\frac{\alpha\lambda\phi}{8t\beta}$, or $\frac{\alpha\lambda\phi}{8t\beta}<\tau<\tau_4$ and $r<r_1$, ↓; otherwise, ↑	↑
	D^{s*}	π_p^{s*}	π_c^{s*}	π_m^{s*}
r	↓	If $\tau\le\tau_5$, ↓; otherwise, ↑	↓	↓

raises ride prices. When the operating cost coefficient is moderate, the platform's response to r varies: If r is low, it lowers the ride price; otherwise, the ride price rises with r.

Regardless of the platform's pricing strategy, passenger demand falls as the matching probability rises. This indicates that introducing AVs enhances passenger–AV stickiness and boosts the platform's market appeal. Therefore, the sensitivity of the platform profits to matching probability is closely tied to the presence of AVs, underscoring their profound influence on the dynamics of the ride-hailing market. When the operating cost coefficient is low, a sharp drop in the quantity of AVs can substantially reduce platform profits. Conversely, when the operating cost coefficient is not restrictive, the platform benefits from the marginal gains of higher ride prices. This finding underscores the need for managers to consider AV positioning in the market when coordinating the matching of HDs and AVs with passengers. In essence, the operating cost of AVs is a pivotal determinant of platform profitability.

Corollary 1. *Platform profits decrease when an increase in r induces a reduction in ride prices.*

Given that the platform's profits depend entirely on ride-hailing services, Proposition 3 shows that passenger demand consistently falls as the matching probability r increases. Consequently, platform profits decline when ride prices are reduced. This pricing adjustment is necessary to mitigate passenger losses caused by a reduction in AV deployment, even at the expense of platform profitability. Furthermore, this result underscores that in the early stages of AV adoption—when the operating cost coefficient is high—platforms must still rely on HDs to meet passenger demand. During this phase, platforms may raise ride prices prudently to ensure and sustain profitability. This nuanced dynamic reflects the delicate balance platforms must strike among adapting to technological change, managing passenger demand, and maintaining financial viability.

Proposition 4. *The impacts of an increase in the operating cost coefficient τ on the equilibrium outcomes are presented in vTable 8.*

Proposition 4 shows that a high operating cost coefficient presents a substantial obstacle to the adoption of AVs in ride-hailing services. First, an increase in the operating cost coefficient raises the cost burden of operating AVs for the joint company. In response, the joint company reduces the size of its AV fleet. Notably, the manufacturer does not raise the OEM price to pursue greater marginal profits. This is because the joint company's decision significantly reduces the manufacturer's overall revenue, prompting it to lower the OEM price to slow the decline in AV deployment. Unexpectedly, the platform did not raise ride prices to attract more HDs and offset the reduction in the scale of AVs. Instead, it lowers ride prices to retain passengers who might leave due to the reduction in AVs. The rationale is that passengers are more sensitive to ride prices, and raising prices amid a reduction in AV supply could accelerate attrition, thereby exacerbating platform profit losses. Therefore, a key role of the platform in the AV segment is to reduce associated costs, ultimately benefits all stakeholders in the ride-hailing ecosystem.

Proposition 5. *The impacts of an increase in the investment ratio ϕ on the equilibrium outcomes are shown in Table 9.*

Investments from both the platform and manufacturer in the joint company serve as the driving force behind the growth of the autonomous driving business. An increase in the investment ratio inevitably incentivizes the joint company to expand the AV fleet in the market. Interestingly, an increase in the investment ratio ϕ seems to reduce manufacturer profits. However, our findings indicate that an increase in ϕ actually leads to higher profit for the manufacturer. This occurs because the manufacturer passes investment costs onto the OEM price, thereby aiming for higher marginal profits. Moreover, the higher investment ratio motivates the joint company to place more orders with the manufacturer, ultimately generating greater returns for the manufacturer under high-investment scenarios. In ride-hailing services, an increase in AV numbers enhances the platform's appeal to passengers, thereby boosting demand. Capitalizing on this opportunity, the platform raises ride prices to secure higher marginal profits, which in turn increases the joint company's profitability. Furthermore, higher ride prices incentivize more HDs to join the platform, further enhancing its service capacity.

Nevertheless, the platform cannot consistently sustain elevated profitability in the context of rising ride prices and growing passenger demand; instead, its profitability is constrained by the operating cost coefficient and investment ratio. As illustrated in Figure 3a,b, the platform realizes enhanced returns from increased investments exclusively when the operating cost coefficient is relatively low, or when a moderate operating cost coefficient coincides with a high investment ratio. Conversely, in alternative scenarios, the platform may enter a "burning money" mode (as shown in Figure 3d), exemplified by Uber Autopilot, which consumed $2.5 billion in 5 years.[16] When operating costs fall within the middle range, the platform faces the imperative of balancing inputs and outputs. In such scenarios, only higher investments can effectively stimulate AV while concomitantly generating the expected returns. The platform needs to be particularly cautious when investing in AVs. Specifically, under moderate operating costs, the platform should avoid adopting excessively high investment ratios in its collaboration with the manufacturer (as shown in Figure 3c), constituting a salient finding of this study. Although elevated investment ratios may increase revenue in the platform's ride-hailing business, they fail to offset the disproportionately high investment expenditures, thereby resulting in diminished profitability.

TABLE 8 | The impacts of increased τ on the equilibrium outcomes.

	q^{s*}	w^{s*}	p^{s*}	k^{s*}	D^{s*}	π_p^{s*}	π_c^{s*}	π_m^{s*}
τ	↓	↓	↓	↓	↓	↓	↓	↓

4.3 | Strategy *d*

This subsection explores a differentiated pricing strategy that distinguishes the ride price for AVs from that charged for HDs. Under strategy d, the profit functions of the platform, joint company, and manufacturer are outlined as follows:

$$\pi_p^d = (1-\beta)rpD^d + (1-\phi)(1-r)\delta pD^d \tag{4}$$

$$\pi_c^d = \phi(1-r)\delta pD^d + \phi wq - wq - \tau q^2 \tag{5}$$

$$\pi_m^d = (1-\phi)wq \tag{6}$$

Similar to Section 4.2, the profit maximization problem is resolved through backward induction, and the equilibrium outcomes are presented in Table 10.

Here, to ensure positive equilibrium outcomes, it is essential that $\tau > \widetilde{\tau} = \frac{\alpha^2\delta\lambda\phi(1-r)}{4(\delta(1-r)+r(1-t\alpha\beta))}$ and $\delta < \overline{\delta}$. This prevents an excessive disparity between the pricing standards for AVs and HDs, ensuring that HDs maintain sufficient activity to provide services to passengers. Under the strategy d, we complement the results that were not identified under strategies n and s.

Proposition 6. *The impacts of an increase in the network externality coefficient α on the equilibrium outcomes are illustrated in Table 11.*

When the application of AVs is accompanied by a differentiated pricing strategy, the traditionally monotonic influence of passengers' network externalities on the partial equilibrium outcomes undergoes a non-monotonic shift. This deviation stems from the effect of the differentiated pricing ratio on the dynamic interaction between ride prices and the quantity of AVs deployed in the market. When passengers exhibit heightened concerns regarding vehicle availability on the platform, coupled with a high differentiated pricing ratio and a substantial operating cost coefficient, a noteworthy scenario emerges. The quantity of AVs in the market decreases as the network externality increases. The manufacturer, as a strategic measure, elects to lower the OEM price, thereby accepting a temporary reduction in profits. This decision aims to catalyze growth in the autonomous driving sector. Interestingly, this outcome defies the traditional assumption that heightened passenger concerns over vehicle supply would typically encourage the platform to maximize vehicle availability to satisfy demand. Instead, the manufacturer's decision reflects a forward-looking strategy that prioritizes long-term industry development over short-term profitability. However, under strategy d, an excessively high differentiated pricing ratio combined with a substantial operating cost coefficient imposes prohibitive

TABLE 9 | The impacts of increased ϕ on the equilibrium outcomes.

	q^{s*}	w^{s*}	p^{s*}	k^{s*}	D^{s*}	π_p^{s*}	π_c^{s*}	π_m^{s*}
ϕ	↑	↑	↑	↑	↑	If $\tau \le \tau_6$, or $\tau_6 < \tau < \tau_7$ and $\phi \ge \phi_1$, or $\tau_7 \le \tau < \frac{8}{7}\tau_6$ and $\phi_1 \le \phi \le \phi_2$, ↑; otherwise, ↓	↑	↑

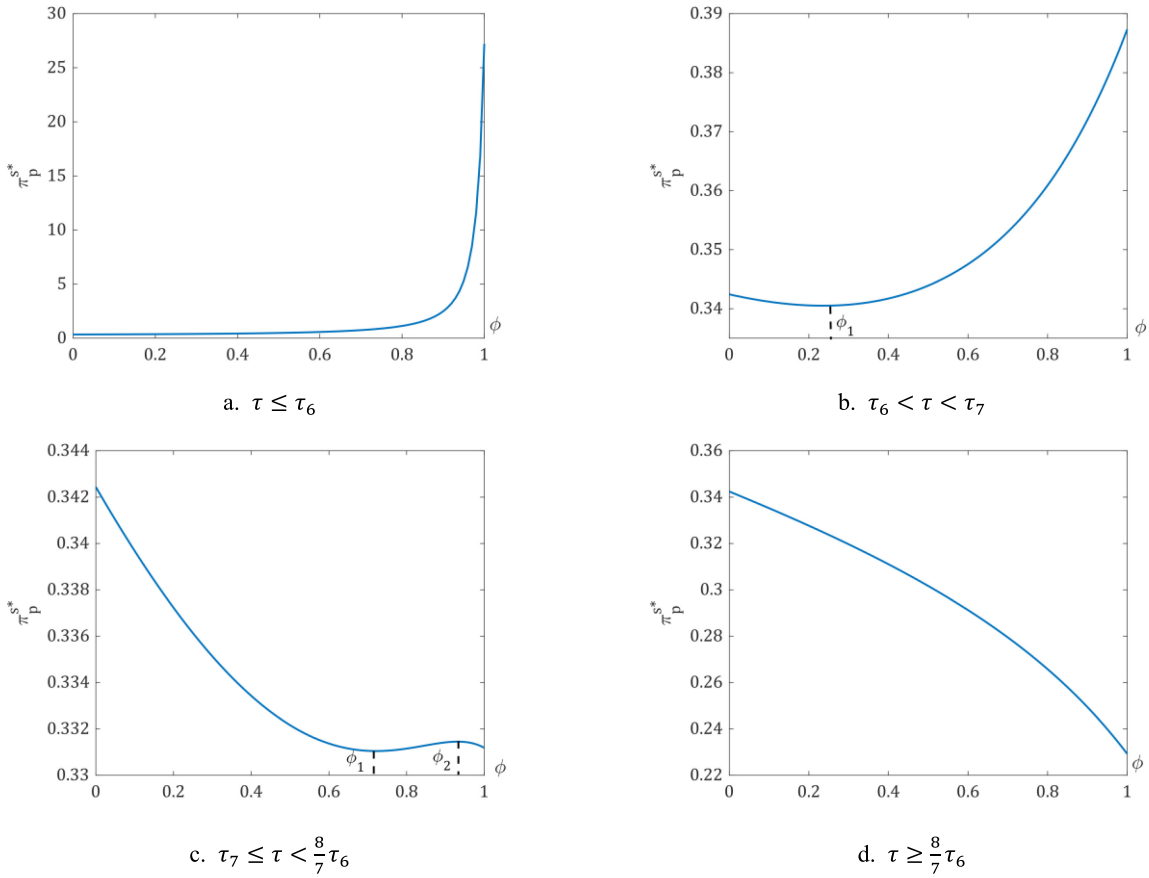


FIGURE 3 | Impact of the investment ratio ϕ on the platform's profits. $\alpha = 0.2$, $t = 6$, $\beta = 0.7$, $\lambda = 1.2$, $r = 0.6$, and $f = 0.06$.

TABLE 10 | The optimal outcomes under the strategy d.

Strategy d			
w^{d*}	$\frac{\alpha\delta\lambda\phi(1-r)(1-\alpha f)}{4\Lambda(1-\phi)}$	D^{d*}	$\frac{\lambda(1-\alpha f)\left(8\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}{4\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}$
q^{d*}	$\frac{\alpha\delta\lambda\phi(1-r)(1-\alpha f)}{2\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}$	π_p^{d*}	$\frac{\lambda(\delta(1-\phi)(1-r)+r(1-\beta))(1-\alpha f)^2\left(8\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)^2}{16\Lambda\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)^2}$
p^{d*}	$\frac{(1-\alpha f)\left(8\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}{4\Lambda\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}$	π_c^{d*}	$\frac{\delta\lambda\phi(1-r)(1-\alpha f)^2\left(16\Lambda\tau-3\alpha^2\delta\lambda\phi(1-r)\right)}{16\Lambda\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}$
k^{d*}	$\frac{rt\beta(1-\alpha f)\left(8\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}{4\Lambda\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}-f$	π_m^{d*}	$\frac{(\alpha\delta\lambda\phi(1-r)(1-\alpha f))^2}{8\Lambda\left(4\Lambda\tau-\alpha^2\delta\lambda\phi(1-r)\right)}$

Note: Here, $\Lambda = \delta(1-r) + r(1-t\alpha\beta)$.

costs on both passengers and the autonomous driving business. Under such conditions, a high network externality for passengers compels the joint company to reduce the AV deployment. Concurrently, the platform mitigates losses by lowering ride prices to retain passengers, leading to a decline in HD supply.

Proposition 7. *The impacts of an increase in the differentiated pricing ratio δ on the equilibrium outcomes are presented in Table 12.*

The differentiated pricing ratio constitutes a key parameter that evaluates the pricing distinction between services provided by HDs and AVs. Under both discounted and premium pricing strategies, as δ increases, the platform reduces the base ride price to prevent passengers from facing excessive costs when choosing AVs. Nevertheless, the platform's capital contributions to the joint company from the platform continue to increase. As a result, the joint company expands its autonomous driving business, while the manufacturer enhances profitability by raising the OEM price. However, although a lower base

ride price increases passenger surplus, it undermines HDs' earnings. Therefore, a higher degree of differentiated pricing does not necessarily ensure sustained profitability for the platform. The platform benefits from higher differentiated pricing ratio only when the operating cost coefficient is relatively low, primarily due to the market expansion effect generated by a larger AV fleet. Additionally, the findings of Proposition 7 suggest that the interests of decision-makers in the ride-hailing market may diverge following the introduction of AVs. Section 5 further examines the platform's strategic decisions regarding the autonomous driving business.

5 | Strategic Analysis

This section examines the implications of introducing AVs into the ride-hailing market. It further investigates the conditions under which platforms adopt AVs by comparing the optimal decisions and profits associated with three strategies for each decision-maker. The analysis evaluates the market benefits of each pricing strategy, considering the profits of respective decision-makers.

Proposition 8 initiates this analysis by comparing the ride price, HD supply, and passenger demand before and after the introduction of AVs.

Proposition 8. *(1) When* $\tau < min\left\{\frac{(1-rt\alpha\beta+(1-r)t\alpha\beta)\alpha\lambda\phi}{8t\beta(1-rt\alpha\beta)}, \frac{\alpha^2\delta\lambda\phi((1-t\alpha\beta)(1-2r)-2\delta(1-r))}{8(\delta(1-r)+r(1-t\alpha\beta))(1-t\alpha\beta-\delta)}\right\}$, $p^{i*} > p^{n*}$; *when* $\tau \geq max\left\{\frac{(1-rt\alpha\beta+(1-r)t\alpha\beta)\alpha\lambda\phi}{8t\beta(1-rt\alpha\beta)}, \frac{(1-rt\alpha\beta+(1-r)t\alpha\beta)\alpha\lambda\phi}{8t\beta(1-rt\alpha\beta)}\right\}$, $p^{i*} \leq p^{n*}$; *(2)* $k^{i*} < k^{n*}$; *(3)* $D^{i*} > D^{n*}$. *Here,* $i \in \{s, d\}$.

Proposition 8 reveals that a lower operating cost coefficient incentivizes the joint company to deploy a larger number of AVs to the market. This expansion strengthens both passenger and platform stickiness, thereby motivating the platform to raise ride prices. However, when the operating cost coefficient exceeds a certain threshold, the joint company can afford to deploy only a limited number of AVs. In response, the platform lowers ride prices to retain passengers. Regardless of whether the platform adopts the same or differentiated pricing strategy, introducing AVs leads to the exit of a portion of HDs from the market. This outcome aligns with the platform's primary objective of reducing wage costs associated with HD labor. Conversely, either expanding the fleet size or reducing ride prices emerges as a pivotal factor influencing passengers to choose the platform. In this respect, the strategy ultimately benefits platform passengers.

Proposition 9. *The platform is motivated to introduce AVs when* $max\{\overline{\tau}, \widetilde{\tau}\} < \tau \leq min\{\tau_{12}, \tau_{13}\}$.

Proposition 9 gives a condition for the platform to implement the AVs introduction strategy: The operating costs associated with AVs must remain within an economically sustainable range. As established in Proposition 4, a lower operating cost coefficient correlates with higher ride prices, greater passenger demand, and enhanced platform profitability. Proposition 8 reinforces this finding, demonstrating that AV introduction is most advantageous when the operating cost coefficient is low, thereby optimizing both marginal and market effects. Consequently, under these conditions, the platform can achieve greater profitability through higher ride prices and increased passenger demand. Therefore, in practical strategy planning, platforms and manufacturers must be prepared to bear the associated costs required to meet a specific threshold (i.e., $\tau > max\{\overline{\tau}, \widetilde{\tau}\}$) when jointly entering the autonomous driving market. Simultaneously, platforms should prioritize partnerships with manufacturers possessing established expertise in autonomous driving technology, thereby avoiding excessive costs on technology R&D and data processing (i.e., $\tau \leq min\{\tau_{12}, \tau_{13}\}$). This explains why numerous platforms collaborate successfully with established manufacturers, such as Didi Chuxing's strategic partnership with Guangzhou Automobile Group's Aion, and Uber's and Lyft's collaborations with Waymo for driverless ride-hailing services.

Proposition 10. *(1) When the discounted pricing strategy is adopted (i.e.,* $\delta < 1$*),* $p^{d*} > p^{s*}$, $w^{d*} < w^{s*}$, $q^{d*} < q^{s*}$, $k^{d*} > k^{s*}$, $D^{d*} < D^{s*}$, $\pi_c^{d*} < \pi_c^{s*}$, *and* $\pi_m^{d*} < \pi_m^{s*}$; *when the premium pricing strategy is adopted (i.e.,* $\delta > 1$*),* $p^{d*} < p^{s*}$, $w^{d*} > w^{s*}$, $q^{d*} > q^{s*}$, $k^{d*} < k^{s*}$, $D^{d*} > D^{s*}$, $\pi_c^{d*} > \pi_c^{s*}$, *and* $\pi_m^{d*} > \pi_m^{s*}$;

(2) If $max\{\overline{\tau}, \widetilde{\tau}\} < \tau \leq \tau_{11}$, *when the discounted pricing strategy is adopted (i.e.,* $\delta < 1$*),* $\pi_p^{d*} < \pi_p^{s*}$; *when the premium pricing strategy is adopted (i.e.,* $\delta > 1$*),* $\pi_p^{d*} > \pi_p^{s*}$. *If* $\tau_{11} < \tau \leq min\{\tau_{12}, \tau_{13}\}$, *when the discounted pricing strategy is adopted (i.e.,* $\delta < 1$*),* $\pi_p^{d*} > \pi_p^{s*}$; *when the premium pricing strategy is adopted (i.e.,* $\delta > 1$*),* $\pi_p^{d*} < \pi_p^{s*}$.

TABLE 12 | The impacts of increased δ on equilibrium outcomes.

	q^{d*}	w^{d*}	p^{d*}	k^{d*}	D^{d*}	π_p^{d*}	π_c^{d*}	π_m^{d*}
δ	↑	↑	↓	↓	↑	If $\widetilde{\tau} < \tau \leq \tau_{11}$, ↑; otherwise, ↓	↑	↑

TABLE 11 | The impacts of increased α on equilibrium outcomes.

	q^{d*}	w^{d*}	p^{d*}	k^{d*}
α	If $\alpha > \frac{1}{2f}$, $\delta > \frac{r(1-\alpha f(2-t\alpha\beta))}{(1-r)(2\alpha f-1)}$ and $\tau > \tau_8$, ↓; otherwise, ↑	If $\alpha > \frac{1}{2f}$ and $\delta > \frac{r(1-\alpha f(2-t\alpha\beta))}{(1-r)(2\alpha f-1)}$, ↓; otherwise, ↑	If $\delta > \frac{r(t\beta-f)}{f(1-r)}$ and $\tau > \tau_9$, ↓; otherwise, ↑	If $\delta > \frac{r(t\beta-f)}{f(1-r)}$ and $\tau > \tau_9$, ↓; otherwise, ↑
		π_m^{d*}		
α		If $\alpha > \frac{1}{2f}$, $\delta > \frac{r(1-\alpha f(2-t\alpha\beta))}{(1-r)(2\alpha f-1)}$ and $\tau > \tau_{10}$, ↓; otherwise, ↑		

Corollary 2. *If $max\{\overline{\tau}, \widetilde{\tau}\} < \tau \leq \tau_{11}$, implementing a premium pricing strategy can achieve a win–win–win situation for the platform, manufacturer, and joint company.*

Once the platform decides to adopt the strategy of introducing AVs under specific conditions, it must further determine the optimal pricing strategy. Proposition 10 compares the equilibrium outcomes under the same pricing and differentiated pricing strategies. First, as shown in Corollary 2, when the operating cost coefficient is low (i.e., $max\{\overline{\tau}, \widetilde{\tau}\} < \tau \leq \tau_{11}$), adopting a premium pricing strategy yields mutually beneficial outcomes for the platform, joint company, and manufacturer. However, this strategy entails reducing the base ride price for HDs, which may reduce HD earnings and prompt drivers with high-income expectations to exit the market. Conversely, under these conditions, a discounted pricing strategy may reduce passenger demand due to higher base ride prices and a smaller fleet size, thereby making the nominally discounted pricing strategy less effective in enhancing consumer surplus. Second, when the operating cost coefficient is relatively high (i.e., $\tau_{11} < \tau \leq min\{\tau_{12}, \tau_{13}\}$), the joint company is reluctant to deploy AVs on a large scale due to cost considerations. In such cases, the platform adopts a discounted pricing strategy to attract passengers, enabling both the platform and HDs to realize higher marginal returns. However, the joint company reduces AV deployment, resulting in profit losses for the manufacturer. Under such circumstances, a conflict of interest emerges between the platform and the joint company/manufacturer. From the perspective of fostering long-term cooperation with the manufacturer, the same pricing strategy serves as a compromise choice for the platform. To achieve a genuine win–win outcome for the platform, joint company, and manufacturer, the optimal approach is to collaboratively advance autonomous driving technology, thereby reducing operating costs and enabling the adoption of a premium pricing strategy. It is important to note that the heightened probability of successful passenger-HD matching, coupled with a limited scope of the autonomous driving business, raises the platform's acceptable threshold for the operating cost coefficient. This higher threshold is more conducive to implementing a premium pricing strategy, thereby increasing the likelihood of achieving mutually beneficial outcome for all decision-makers involved, as illustrated in Figure 4.

6 | Extension

In the extension, we consider four scenarios that exist in reality: (1) government subsidies for R&D and operations; (2) HD same-side network externalities; (3) heterogeneous investment ratios between the platform and the manufacturer; and (4) passengers' sentiment towards AVs. All equilibrium outcomes from the extended models are provided in the Supporting Information.

6.1 | Considering Government Subsidies for R&D and Operations

In practice, when firms invest in developing emerging technologies, governments frequently provide R&D subsidies. For example, in China, policies such as special funds, tax incentives, and research project support (e.g., the National Key R&D Program) are actively promoting the development of AVs. Therefore, in this section, we consider a scenario in which the government provides a certain proportion of subsidies for the R&D and operational activities undertaken by the joint company engaged in AV development. Following the related literature (Niu, Zhang, et al. 2024), we assume that the government subsidizes a proportion η of the R&D and operating costs. Accordingly, the cost borne by the joint company is expressed as $(1-\eta)\tau q^2$. We derive all equilibrium outcomes under different pricing strategies adopted by the platform in the presence of government subsidies. Building on the basic model and considering the conflicting interests of decision-makers in the ride-hailing market under different strategies, we assess both pricing strategies from a social welfare perspective. Total social welfare (denoted) comprises the profits of all decision-makers, the surplus of drivers and passengers, and the government's subsidy expenditure. Based on this, we conduct numerical simulations and present the findings in Observation 1.

Observation 1. *The premium pricing strategy enhances overall social welfare only when the operating cost coefficient is sufficiently low; otherwise, the discounted pricing strategy is more effective.*

As illustrated in Figure 5, premium pricing yields higher social welfare only under a very low operating cost coefficient. A comparison between Figures 4 and 5 further reveals that the threshold of the operating cost coefficient affecting social welfare is significantly lower than that affecting platform profitability. Within the interval between these two thresholds, the platform, joint company, manufacturer, and passengers achieve greater benefits under the premium pricing strategy; however, the discounted pricing strategy still generates higher social welfare. This finding suggests an important implication for decision-makers in the ride-hailing market: Even if platforms introduce AVs to stimulate market demand and reduce commission payments to HDs, they must remain attentive to the interests of drivers while pursuing profitability. Moreover, Figure 5 shows that the probability of attaining a win–win outcome increases with the level of R&D subsidies. In this scenario, the platform, manufacturer, joint company, and society as a whole all benefit under the premium pricing strategy. In contrast, when the operating cost coefficient is high, the discounted pricing strategy enables the platform and drivers to jointly generate greater social welfare.

6.2 | Considering Same-Side Network Externalities for HDs

When AVs are introduced to the platform, a competitive relationship arises between AVs and HDs. This competition encroaches on the original HD supply market, producing a cannibalization effect. For example, Freund et al. (2022) and Castro and Frazelle (2024) examine the competition between HDs and AVs in their studies. Moreover, evidence shows that HDs may engage in collusion during operations to maintain a low adoption rate of AVs (Bai, Heese, et al. 2023; Tripathy et al. 2023). In this section, we investigate the same-side network externalities for HDs to illustrate their competitive exclusion of AVs. In this scenario, the utility of an HD is represented as $U_d = rt\beta p - \mu q - f - \varepsilon$, where

μ is the same-side network externality coefficient. Accordingly, the final number of active HDs, denoted as $k = rt\beta p - \mu q - f$, can be derived.

Proposition 11. *The impacts of an increase in the same-side network externality coefficient μ on the equilibrium outcomes are illustrated in Table 13.*

Here, $i \in \{s, d\}$.

Proposition 11 examines how the same-side network externality coefficient μ for HDs affects equilibrium outcomes. Notably, μ exerts a consistent influence under both pricing strategies. As μ strengthens, the joint company is forced to reduce the quantity of AVs deployed to ensure an adequate supply of HDs. This decision triggers a cascade effect, prompting the manufacturer to decrease the OEM prices. Consequently, the platform lowers ride prices to ease passenger dissatisfaction from the diminished vehicle supply, though passenger demand still declines overall. When μ exceeds a certain threshold, HD utility increases, suggesting that HDs' resistance to AVs provides them with a relative advantage. This occurs as the reduction in the quantity of AVs surpasses the rate of ride price reductions. As a result, the platform, joint company, and manufacturer collectively bear the financial loss resulting from market downsizing.

Next, we explore the impact of the same-side network externality coefficient on the platform's strategic decisions.

Proposition 12. *(1) The platform is incentivized to introduce AVs when $max\{(1-\mu)^2\overline{\tau}, (1-\mu)^2\widetilde{\tau}\} < \tau \le min\{(1-\mu)^2\tau_{12}, (1-\mu)^2\tau_{13}\}$;*

(2) When the discounted pricing strategy is adopted (i.e., $\delta < 1$), $\pi_c^{d*} < \pi_c^{s*}$ and $\pi_m^{d*} < \pi_m^{s*}$; when the premium pricing strategy is adopted (i.e., $\delta > 1$), $\pi_c^{d*} > \pi_c^{s*}$ and $\pi_m^{d*} > \pi_m^{s*}$;

(3) If $max\{(1-\mu)^2\overline{\tau}, (1-\mu)^2\widetilde{\tau}\} < \tau \le (1-\mu)^2\tau_{11}$, when the discounted pricing strategy is adopted (i.e., $\delta < 1$), $\pi_p^{d*} < \pi_p^{s*}$; when the premium pricing strategy is adopted (i.e., $\delta > 1$), $\pi_p^{d*} > \pi_p^{s*}$. If $(1-\mu)^2\tau_{11} < \tau \le min\{(1-\mu)^2\tau_{12}, (1-\mu)^2\tau_{13}\}$, when the discounted pricing strategy is adopted (i.e., $\delta < 1$), $\pi_p^{d*} > \pi_p^{s*}$; when the premium pricing strategy is adopted (i.e., $\delta > 1$), $\pi_p^{d*} < \pi_p^{s*}$.

Proposition 12 aligns closely with the findings of Propositions 9 and 10. Specifically, when considering the same-side network externality coefficient that reflects HDs' attitudes towards AVs, the platform's strategic decision is consistent with the basic model. The difference is that the operating cost coefficient must fall within a narrower threshold range in this scenario. Introducing AVs into the market challenges the collaboration between the platform and manufacturer due to HDs' unfavorable attitudes. Moreover, as the same-side network externality coefficient increases, achieving a win–win–win scenario for the platform, joint company, and manufacturer becomes increasingly difficult. Specifically, the operating cost coefficient required to achieve this mutual benefit decreases at the rate of

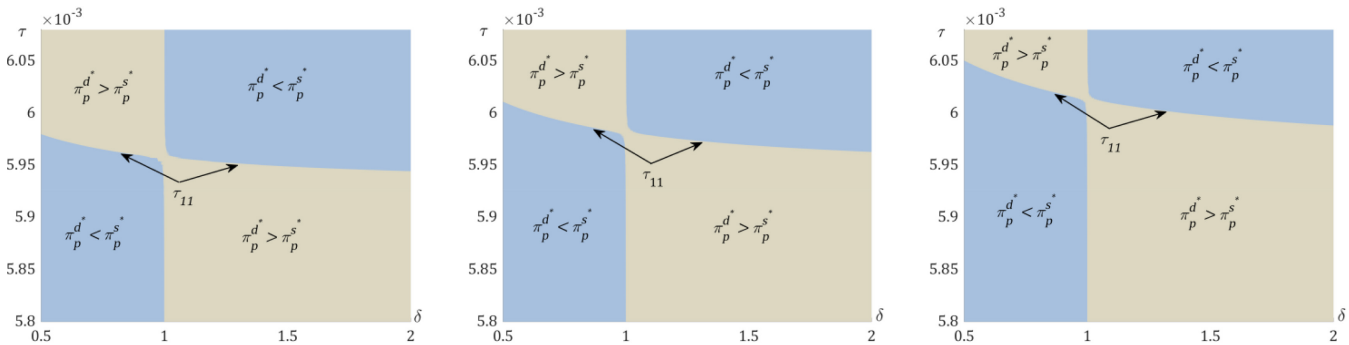


FIGURE 4 | Illustration of (2) in Proposition 10. $\alpha = 0.2$, $t = 6$, $\beta = 0.8$, $\lambda = 1.2$, $\phi = 0.4$, and $f = 0.06$.

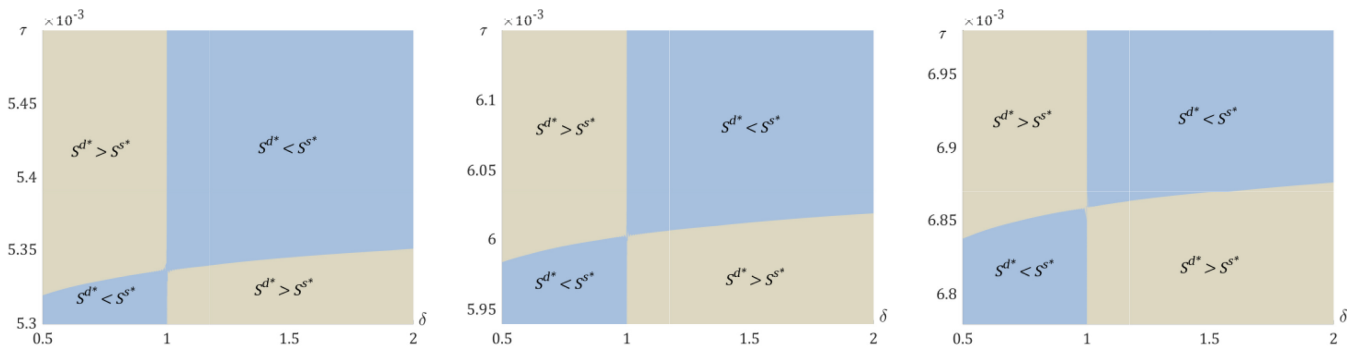


FIGURE 5 | Illustration of Observation 1. $\alpha = 0.2$, $t = 6$, $\beta = 0.8$, $\lambda = 1.2$, $r = 0.3$, $\phi = 0.4$, and $f = 0.06$.

$(1-\mu)^2$. To successfully introduce AVs and ensure the long-term and stable development of the ride-hailing market, platforms must implement effective measures to alleviate HDs' negative attitudes, such as moderately raising commission rates or offering bonuses to HDs (Bai, Heese, et al. 2023; Tripathy et al. 2023).

6.3 | Considering the Heterogeneity of the Investment Ratio Between the Platform and the Manufacturer

In the basic model, the platform and manufacturer allocate a portion of the profits from the AVs business to the joint company to ensure its proper operation, assuming that both parties contribute equally. However, in practice, the profitability of each decision-maker often differs, making it difficult to maintain a consistent investment ratio among investors. Therefore, this subsection explores the scenario where the investment ratios of the platform and manufacturer are heterogeneous. We denote the platform's investment ratio as ϕ_p and the manufacturer's as ϕ_m. All other assumptions remain unchanged from the basic model.

When the investment ratios of the platform and manufacturer are heterogeneous, the equilibrium outcomes under the two pricing strategies reveal that, except for the OEM price of the AVs, the manufacturer's investment ratio does not affect any other equilibrium outcomes. Instead, they depend solely on the platform's investment ratio. From this perspective, an increase in the manufacturer's investment ratio is equivalent to a proportional rise in the OEM price, indicating that the manufacturer's investment has minimal impact on economic efficiency, whereas the platform's investment plays a decisive role. Therefore, the heterogeneity of investment ratios between the platform and manufacturer does not change the results of the basic model regarding the introduction of the AV strategy and the pricing strategy. Moreover, as Proposition 5 highlights, increasing the platform's investment ratio benefits multiple entities in the ride-hailing market, although it does not necessarily lead to higher platform profits. Figure 6 illustrates the effect of the platform's investment ratio on the choice of pricing strategy. Higher platform investment in AVs enhances the affordability of the joint company concerning costs, thereby supporting the autonomous driving business. Consequently, this decision increases the likelihood that the platform, joint company, and manufacturer will achieve a mutually beneficial outcome under a premium pricing strategy.

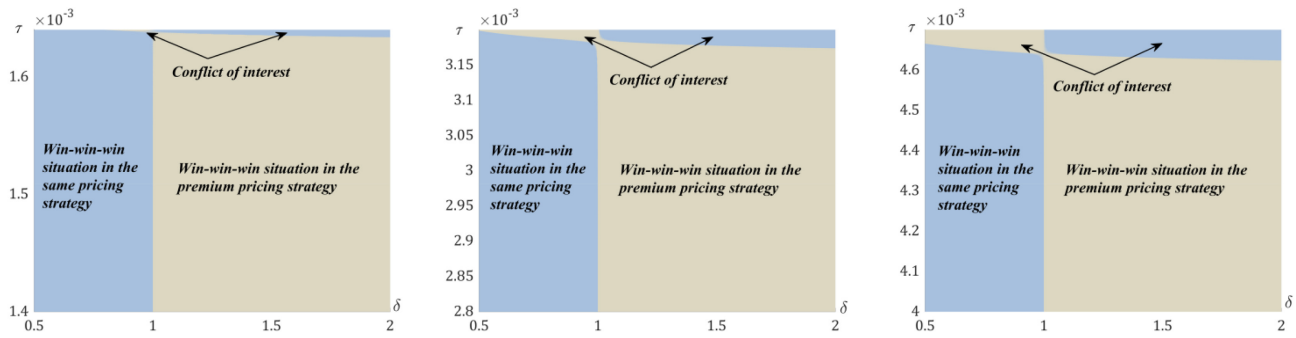
TABLE 13 | The impacts of increased same-side network externality coefficient μ on equilibrium outcomes.

	q^{i*}	w^{i*}	p^{i*}	k^{i*}	D^{i*}	π_p^{i*}	π_c^{i*}	π_m^{i*}
μ	↓	↓	↓	If $\mu < \mu^i$, ↓; otherwise, ↑	↓	↓	↓	↓

6.4 | Considering Passengers' Sentiment Towards AVs

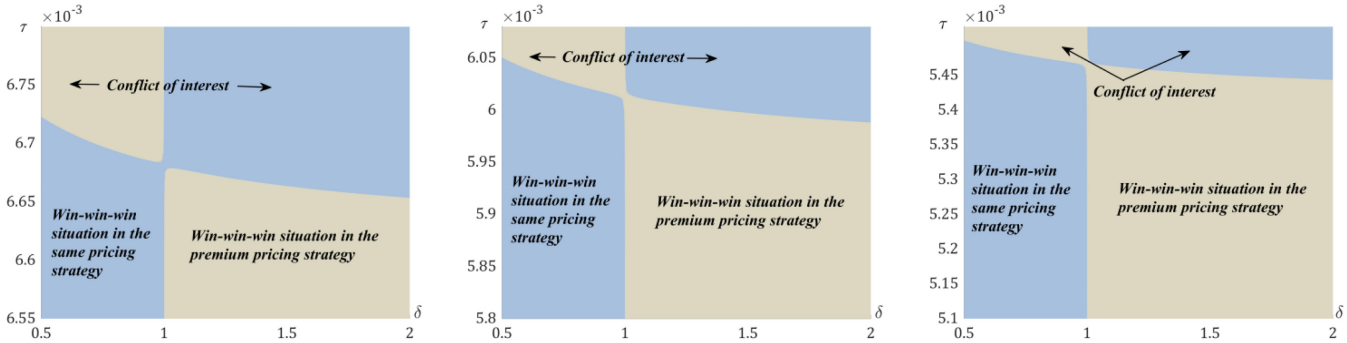
The introduction of AVs not only affects vehicle supply and ride prices but also changes in passengers' perceived value of the platform. For example, conservative passengers may have safety concerns and be unable to fully trust AVs, or they may prefer the opportunity to interact with the HD. In such cases, the perceived value of the platform may decrease for these passengers. Conversely, open-minded passengers may appreciate the novel experience that AVs offer, increasing their perceived value to the platform. Considering passengers' sentiment towards AVs, the initial perceived value of the platform after AV introduction is denoted as θv, where θ represents passengers' acceptance degree of AVs. If $0 < \theta < 1$, passengers are considered to dislike AVs; if $\theta > 1$, they are considered to favor AVs. When the platform adopts the same pricing strategy, a passenger's utility is given by $U_c^s = \theta v - p + \alpha(k+q)$, and passenger demand can be obtained as $D^s = \lambda\left(1 - \frac{p-\alpha(k+q)}{\theta}\right)$. Under a differentiated pricing strategy, a passenger's utility is $U_c^d = \theta v - (rp + (1-r)\delta p) + \alpha(k+q)$, yielding passenger demand $D^d = \lambda\left(1 - \frac{(rp+(1-r)\delta p)-\alpha(k+q)}{\theta}\right)$. Other model setups remain consistent with the basic model.

Analyzing the equilibrium outcomes while accounting for passengers' sentiment towards AVs yields conclusions consistent with the basic model. Specifically, if passengers exhibit a stronger aversion to AVs, the platform's threshold for the operating cost coefficient required to introduce AVs becomes more stringent. This implies that the joint company must invest additional resources to enhance AV safety and intelligence, thereby



a. $\phi_p = 0.1$ b. $\phi_p = 0.2$ c. $\phi_p = 0.3$

FIGURE 6 | The impact of the platform's investment ratio on the selection of pricing strategy. $\alpha = 0.2$, $t = 6$, $\beta = 0.8$, $\lambda = 1.2$, $\phi = 0.4$, $f = 0.06$, and $r = 0.6$.



a. $\theta = 0.9$ b. $\theta = 1$ c. $\theta = 1.1$

FIGURE 7 | The impact of the passengers' acceptance degree on the selection of pricing strategy. $\alpha = 0.2$, $t = 6$, $\beta = 0.8$, $\lambda = 1.2$, $\phi = 0.4$, $f = 0.06$, and $r = 0.6$.

improving passengers' acceptance. Furthermore, as passenger acceptance of AVs grows, the platform is likely to raise ride prices and expand AV deployment in the market. This, in turn, attracts more passengers and promotes a win–win–win outcome for the platform, joint company, and manufacturer, as illustrated in Figure 7.

7 | Conclusions

Ride-hailing is among the most flourishing segments of the sharing economy. However, ride-hailing platforms consistently struggle with profitability. Many of these platforms see the introduction of AVs as a strategic move to enhance profitability. Nevertheless, owing to the complex market dynamics, the timing and pricing strategies are crucial for effectively implementing the application of AVs. To address these challenges, we construct a cooperative game model involving a ride-hailing platform, manufacturer, and joint company established by both entities. In this model, the joint company assumes responsibility for the operating costs of AVs and is obligated to pay the manufacturer an OEM price. The platform must make crucial decisions, including whether to introduce AVs and the subsequent pricing strategy, which are all based on considerations of profitability.

The study results underscore the pivotal role of the operating costs of AVs in shaping the impact of changes in the market environment on ride-hailing. For example, the effects of passengers' network externality, the matching probability between passengers and HDs, the investment ratios of the platform and manufacturer, and the differentiated pricing ratios on the platform's profits are all influenced by the operating costs of AVs. It is crucial to highlight that platform profitability from increased investment is feasible only when the operating cost coefficient is low, or when the operating cost coefficient is moderate, and the investment ratio is high. Otherwise, the platform may find itself in a "burning money" mode. In instances of heterogeneous investment ratios between the platform and manufacturer, it becomes apparent that only the platform's investments significantly affect the ride-hailing market. In line with real-world scenarios, high operating costs may lead the platform to reconsider its involvement in the autonomous driving business. Furthermore, the application of AVs effectively increases the stickiness between passengers and the platform, while simultaneously reducing wages for HDs. Notably, commercializing AVs on a large scale for ride-hailing becomes incentive-worthy for the platform only when the operating cost coefficient is low enough to overcome profitability challenges. In conclusion, when the operating cost coefficient is below a certain threshold, opting for a premium pricing strategy is more beneficial for the profitability of the platform, ensuring a win–win–win situation for the platform, joint company, and manufacturer. Otherwise, the platform should adopt a discounted pricing strategy, which will cause a conflict of interest between the platform, joint company, and manufacturer. In this sense, choosing the same pricing strategy would be more favorable for the long-term cooperation between the platform and manufacturer. In terms of overall social welfare, especially when the driver surplus contribution is substantial, a premium pricing strategy is preferred when the operating cost coefficient is low; otherwise, a discounted pricing strategy proves more effective. The extended model explores scenarios considering subsidies for R&D and operations, the same-side network externality for HDs, heterogeneous investment ratios of the platform and manufacturer, and passengers' sentiment toward AVs, validating the stability of the basic model.

Our research provides innovative insights for ride-hailing platforms in managing AV operations. First, when selecting partners, platforms should target companies with cutting-edge autonomous driving technology. By collaborating on technology, platforms can significantly reduce operating costs, thereby accelerating the large-scale adoption of AVs in ride-hailing. This approach not only enhances market competitiveness but also helps establish industry technology standards. Second, we recommend that platforms conduct comprehensive market and technology trend analyses when they make investment decisions in AV operations. Special attention should be paid to the life cycle cost and return on investment of AVs. By leveraging big data and artificial intelligence, platforms can accurately predict market demand and technological advancements, avoiding blind investments and inefficient "burning money" models, thus ensuring efficient resource allocation. Additionally, when setting pricing strategies, platforms must consider not only their profitability but also the level of their technological and market collaboration with manufacturers. Establishing a shared economic ecosystem can lead to mutual

benefits and jointly advance the development and adoption of autonomous driving technology. To overcome profitability challenges, ride-hailing platforms must keenly focus on the pivotal role and essential needs of HDs. As linchpins in vehicle provision and contributors to societal welfare, HDs are indispensable. Platforms should foster a strong sense of engagement and belonging among HDs by offering comprehensive training, robust technical support, and substantial welfare benefits. This strategic approach will not only bolster operational efficiency but also significantly enhance the companies' commitment to social responsibility and their overall image. In this process, the government can facilitate win–win outcomes among market participants and achieve greater social welfare by increasing the proportion of R&D subsidies.

In future research, several potential directions can be explored. First, the car rental business of ride-hailing platforms is not considered in our current study. In practice, many platforms incorporate car rental services to alleviate the cost burden on HDs, which could influence platforms' approach to the autonomous driving business. Second, we have yet to account for government regulations regarding AVs. Government restrictions on the technology and deployment of AVs present potential challenges for platforms. Third, intense competition among ride-hailing platforms is a prevalent reality. Exploring the dynamics of competition among platforms engaging in the autonomous driving business within a multi-entity game model poses a challenging yet important avenue for future research. Finally, our model assumes a convex combination of ride prices for HDs and AVs. Future research on nonconvex combinations or heterogeneous pricing for these two types of services would be highly valuable.

Author Contributions

Bin Li: conceptualization, methodology, software, formal analysis, writing – original draft. **Meng Zhao:** funding acquisition, supervision, writing – review and editing. **Ninghao Chen:** investigation, conceptualization, validation. **Zhitang Li:** writing – review and editing. **Huijie Sun:** writing – review and editing.

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Conflicts of Interest

The authors declare no conflicts of interest.

Data Availability Statement

The data that support the findings of this study are available from the corresponding author upon reasonable request.

Endnotes

[1] https://www.washingtonpost.com/technology/2019/03/29/even-with-ipo-billions-can-uber-lyft-survive-long-enough-replace-their-drivers-with-machines/.

[2] https://www.didiglobal.com/news/newsDetail?id=1023&type=news.

[3] https://www.nextbigfuture.com/2024/07/china-robotaxi-is-much-bigger-than-waymo.html.

[4] http://bj.people.com.cn/n2/2024/0708/c14540-40905476.html.

[5] https://www.axios.com/2023/08/29/cities-testing-self-driving-driverless-taxis-robotaxi-waymo.

[6] https://www.collimator.ai/post/the-state-of-autonomous-vehicles-in-2023.

[7] https://investor.uber.com/news-events/news/press-release-details/2023/Waymo-and-Uber-Partner-to-Bring-Waymos-Autonomous-Driving-Technology-to-the-Uber-Platform/default.aspx.

[8] https://www.media.volvocars.com/cn/zh-cn/media/pressreleases/225403/uber3.

[9] https://www.didiglobal.com/news/newsDetail?id=1027&type=news.

[10] https://www.yicaiglobal.com/news/baidus-robotaxi-app-apollo-go-gathers-steam-with-discounts-despite-complaints.

[11] https://www.gz.gov.cn/ysgz/xwdt/ysdt/content/mpost_8890044.html.

[12] https://waymo.com/blog/2023/05/waymo-and-uber-partner-to-bring-waymos/.

[13] https://www.uber.com/newsroom/waymo-on-uber/.

[14] https://ride.guru/content/resources/driver-payout-take-home#UBER.

[15] https://www.mot.gov.cn/jiaotongyaowen/202212/t20221228_3730275.html.

[16] https://www.theinformation.com/articles/infighting-busywork-missed-warnings-how-uber-wasted-2-5-billion-on-self-driving-cars.

References

Al-Kanj, L., J. Nascimento, and W. B. Powell. 2020. "Approximate Dynamic Programming for Planning a Ride-Hailing System Using Autonomous Fleets of Electric Vehicles." *European Journal of Operational Research* 284, no. 3: 1088–1106. https://doi.org/10.1016/j.ejor.2020.01.033.

Ao, D., Z. Lai, and S. Li. 2024. "Control of Dynamic Ride-Hailing Networks With a Mixed Fleet of Autonomous Vehicles and For-Hire Human Drivers." *Transportation Research Part E: Logistics and Transportation Review* 189: 103680. https://doi.org/10.1016/j.tre.2024.103680.

Badue, C., R. Guidolini, R. V. Carneiro, et al. 2021. "Self-Driving Cars: A Survey." *Expert Systems With Applications* 165: 113816. https://doi.org/10.1016/j.eswa.2020.113816.

Bai, J., H. S. Heese, and M. Tripathy. 2023. "Hiding in Plain Sight: Surge Pricing and Strategic Providers." *Production and Operations Management* 32, no. 12: 3837–3855. https://doi.org/10.1111/poms.14064.

Bai, J., K. C. So, C. S. Tang, X. M. Chen, and H. Wang. 2019. "Coordinating Supply and Demand on an On-Demand Service Platform With Impatient Customers." *Manufacturing & Service Operations Management* 21, no. 3: 556–570. https://doi.org/10.1287/msom.2018.0707.

Bai, J., and C. S. Tang. 2022. "Can Two Competing On-Demand Service Platforms Be Profitable?" *International Journal of Production Economics* 250: 108672. https://doi.org/10.1016/j.ijpe.2022.108672.

Balakrishnan, A., S. Sundaresan, and B. Zhang. 2014. "Browse-and-Switch: Retail-Online Competition Under Value Uncertainty." *Production and Operations Management* 23, no. 7: 1129–1145. https://doi.org/10.1111/poms.12165.

Baron, O., O. Berman, and M. Nourinejad. 2022. "Introducing Autonomous Vehicles: Adoption Patterns and Impacts on Social Welfare." *Manufacturing & Service Operations Management* 24, no. 1: 352–369. https://doi.org/10.1287/msom.2020.0955.

Benjaafar, S., J.-Y. Ding, G. Kong, and T. Taylor. 2022. "Labor Welfare in On-Demand Service Platforms." *Manufacturing & Service Operations Management* 24, no. 1: 110–124. https://doi.org/10.1287/msom.2020.0964.

Benjaafar, S., Z. Wang, and X. Yang. 2025. "The Impact of Automation on Workers When Workers Are Strategic: The Case of Ride-Hailing." *Manufacturing & Service Operations Management* 27, no. 3: 1332–1682. https://doi.org/10.1287/msom.2023.0416.

Bernstein, F., G. A. DeCroix, and N. B. Keskin. 2021. "Competition Between Two-Sided Platforms Under Demand and Supply Congestion Effects." *Manufacturing & Service Operations Management* 23, no. 5: 1043–1061. https://doi.org/10.1287/msom.2020.0866.

Besbes, O., F. Castro, and I. Lobel. 2021. "Surge Pricing and Its Spatial Supply Response." *Management Science* 67, no. 3: 1350–1367. https://doi.org/10.1287/mnsc.2020.3622.

Bimpikis, K., O. Candogan, and D. Saban. 2019. "Spatial Pricing in Ride-Sharing Networks." *Operations Research* 67, no. 3: 744–769. https://doi.org/10.1287/opre.2018.1800.

Biswas, D., H. Jalali, A. H. Ansaripoor, and P. De Giovanni. 2023. "Traceability vs. Sustainability in Supply Chains: The Implications of Blockchain." *European Journal of Operational Research* 305, no. 1: 128–147. https://doi.org/10.1016/j.ejor.2022.05.034.

Cai, Z., D. Mo, W. Tang, Y. Chen, and X. M. Chen. 2023. "A Two-Period Game-Theoretical Model for Heterogeneous Ride-Sourcing Platforms With Asymmetric Competition and Mixed Fleets." *Transportation Research Part E: Logistics and Transportation Review* 178: 103279. https://doi.org/10.1016/j.tre.2023.103279.

Cao, Y., X. Li, F. Kou, and G. Wan. 2025. "The More, the Better? Effects of Same-Side Network Externalities on Customer Loyalty in Ride-Hailing Aggregation Platforms." *Information Systems Frontiers*: 1–17. https://doi.org/10.1007/s10796-025-10593-3.

Castillo, J. C., D. Knoepfle, and E. G. Weyl. 2025. "Matching and Pricing in Ride Hailing: Wild Goose Chases and How to Solve Them." *Management Science* 71, no. 5: 4377–4395. https://doi.org/10.1287/mnsc.2022.00096.

Castro, F., and A. E. Frazelle. 2024. "Getting Out of Your Own Way: Introducing Autonomous Vehicles on a Ride-Hailing Platform." *Production and Operations Management* 33, no. 10: 2014–2030. https://doi.org/10.1177/10591478241264796.

Chen, N., B. Li, M. Zhao, J. Ren, and J. Su. 2025. "Sharing Channel Choices for Electric Vehicle Battery Manufacturers When Considering Range Anxiety Among Heterogeneous Consumers." *Asia Pacific Journal of Marketing and Logistics* 37, no. 4: 948–969. https://doi.org/10.1108/APJML-05-2024-0586.

Choi, T.-M., S. Guo, N. Liu, and X. Shi. 2020. "Optimal Pricing in On-Demand-Service-Platform-Operations With Hired Agents and Risk-Sensitive Customers in the Blockchain Era." *European Journal of Operational Research* 284, no. 3: 1031–1042. https://doi.org/10.1016/j.ejor.2020.01.049.

Dong, T., Z. Xu, Q. Luo, Y. Yin, J. Wang, and J. Ye. 2021. "Optimal Contract Design for Ride-Sourcing Services Under Dual Sourcing." *Transportation Research Part B: Methodological* 146: 289–313. https://doi.org/10.1016/j.trb.2021.01.014.

Dong, Y., Y. Sun, D. Wang, and E. O. D. Waygood. 2025. "From First Ride to Regular User: Understanding the Factors Influencing Continuous Use Intention of Autonomous Taxis." *Transportation Research Part F: Traffic Psychology and Behaviour* 113: 106–123. https://doi.org/10.1016/j.trf.2025.04.019.

Fan, X., K. Chen, and Y.-J. Chen. 2023. "Is Price Commitment a Better Solution to Control Carbon Emissions and Promote Technology Investment?" *Management Science* 69, no. 1: 325–341. https://doi.org/10.1287/mnsc.2022.4365.

Feng, H., N. Feng, L. Zhang, Z. Jiang, and M. Li. 2025. "A Study of Ride-Hailing Platforms' Business Models in the Presence of Surge Demand." *Information Systems Research*. https://doi.org/10.1287/isre.2022.0695.

Feng, S., P. Duan, J. Ke, and H. Yang. 2022. "Coordinating Ride-Sourcing and Public Transport Services With a Reinforcement Learning Approach." *Transportation Research Part C: Emerging Technologies* 138: 103611. https://doi.org/10.1016/j.trc.2022.103611.

Feng, Y., R. Niazadeh, and A. Saberi. 2023. "Two-Stage Stochastic Matching and Pricing With Applications to Ride Hailing." *Operations Research* 72, no. 4: 1574–1594. https://doi.org/10.1287/opre.2022.2398.

Fielbaum, A., D. Salas, R. Zhang, and F. Castro. 2025. "Idle Wage as a Tool to Regulate the Relationship Between Ride-Hailing Platforms and Drivers." *Transportation Research Part C: Emerging Technologies* 174: 105113. https://doi.org/10.1016/j.trc.2025.105113.

Freund, D., I. Lobel, and J. K. Zhao 2022. On the Supply of Autonomous Vehicles in Open Platforms. Available at SSRN 4178508.

Hartman, R. S., D. Wheeler, and M. Singh. 1997. "The Cost of Air Pollution Abatement." *Applied Economics* 29, no. 6: 759–774. https://doi.org/10.1080/000368497326688.

He, P., Q. Shang, W. Pedrycz, and Z.-S. Chen. 2024. "Short Video Creation and Traffic Investment Decision in Social E-Commerce Platforms." *Omega* 128: 103129. https://doi.org/10.1016/j.omega.2024.103129.

Hu, X., S. Zhou, X. Luo, J. Li, and C. Zhang. 2024. "Optimal Pricing Strategy of an On-Demand Platform With Cross-Regional Passengers." *Omega* 122: 102947. https://doi.org/10.1016/j.omega.2023.102947.

Huang, L., B. Liu, and R. Zhang. 2024. "Channel Strategies for Competing Retailers: Whether and When to Introduce Live Stream?" *European Journal of Operational Research* 312, no. 2: 413–426. https://doi.org/10.1016/j.ejor.2023.06.017.

Ji, G., T. Fu, and S. Li. 2023. "Optimal Selling Format Considering Price Discount Strategy in Live-Streaming Commerce." *European Journal of Operational Research* 309, no. 2: 529–544. https://doi.org/10.1016/j.ejor.2023.01.034.

Jiang, Z.-Z., G. Kong, and Y. Zhang. 2021. "Making the Most of Your Regret: Workers' Relocation Decisions in On-Demand Platforms." *Manufacturing & Service Operations Management* 23, no. 3: 695–713. https://doi.org/10.1287/msom.2020.0916.

Ke, J., Z. Zhu, H. Yang, and Q. He. 2021. "Equilibrium Analyses and Operational Designs of a Coupled Market With Substitutive and Complementary Ride-Sourcing Services to Public Transits." *Transportation Research Part E: Logistics and Transportation Review* 148: 102236. https://doi.org/10.1016/j.tre.2021.102236.

Li, B., M. Zhao, Z. Hao, and X. Wang. 2023. "Strategy Selection of Ride-Sourcing Platform Supply Chain Under the Targets of "Double Carbon": Integration or Cooperation." *Computers & Industrial Engineering* 181: 109293. https://doi.org/10.1016/j.cie.2023.109293.

Li, J., J. Ou, and B. Cao. 2023. "The Roles of Cooperative Advertising and Endogenous Online Price Discount in a Dual-Channel Supply Chain." *Computers & Industrial Engineering* 176: 108980. https://doi.org/10.1016/j.cie.2023.108980.

Li, Y., X. Li, and S. Zhang. 2021. "Optimal Pricing of Customized Bus Services and Ride-Sharing Based on a Competitive Game Model." *Omega* 103: 102413. https://doi.org/10.1016/j.omega.2021.102413.

Lian, Z., and G. van Ryzin. 2024. "Capturing the Benefits of Autonomous Vehicles in Ride Hailing: The Role of Market Configuration." *Management Science* 71, no 7: 5491–5510. https://doi.org/10.1287/mnsc.2020.03112.

Lin, X., C. Sun, B. Cao, Y.-W. Zhou, and C. Chen. 2021. "Should Ride-Sharing Platforms Cooperate With Car-Rental Companies? Implications for Consumer Surplus and Driver Surplus." *Omega* 102: 102309. https://doi.org/10.1016/j.omega.2020.102309.

Liu, X., and Q. Zhou. 2024. "Interactions Between Supplier Encroachment Format and Pricing Scheme With Horizontal Shareholding." *Expert Systems With Applications* 256: 124934. https://doi.org/10.1016/j.eswa.2024.124934.

Mo, D., X. M. Chen, and J. Zhang. 2022. "Modeling and Managing Mixed On-Demand Ride Services of Human-Driven Vehicles and Autonomous Vehicles." *Transportation Research Part B: Methodological* 157: 80–119. https://doi.org/10.1016/j.trb.2022.01.003.

Mo, D., J. Yu, and X. M. Chen. 2020. "Modeling and Managing Heterogeneous Ride-Sourcing Platforms With Government Subsidies on Electric Vehicles." *Transportation Research Part B: Methodological* 139: 447–472. https://doi.org/10.1016/j.trb.2020.07.006.

Niu, B., F. Xie, and L. Chen. 2024. "Analysis of Global Brands' Green Procurement Strategy: Co-Opetitive Local Sourcing vs. Overseas Sourcing Under Delivery Disruption." *Omega* 123: 102991. https://doi.org/10.1016/j.omega.2023.102991.

Niu, B., N. Zhang, and J. Zhang. 2024. "How to Induce Multinational Firms' Local Sourcing to Break Carbon Lock-In?" *European Journal of Operational Research* 315, no. 2: 613–625. https://doi.org/10.1016/j.ejor.2023.12.009.

Nourinejad, M., and M. Ramezani. 2020. "Ride-Sourcing Modeling and Pricing in Non-Equilibrium Two-Sided Markets." *Transportation Research Part B: Methodological* 132: 340–357. https://doi.org/10.1016/j.trb.2019.05.019.

Ren, D., R. Guo, Y. Lan, and C. Shang. 2021. "Shareholding Strategies for Selling Green Products on Online Platforms in a Two-Echelon Supply Chain." *Transportation Research Part E: Logistics and Transportation Review* 149: 102261. https://doi.org/10.1016/j.tre.2021.102261.

Sharma, P., and C. Rana. 2024. "Artificial Intelligence Based Object Detection and Traffic Prediction by Autonomous Vehicles—A Review." *Expert Systems With Applications* 255: 124664.

Shi, B., L. Fu, Z. Cao, and L. Zhu. 2024. "A Second-Pricing Based Incentive-Compatible Mechanism for Matching and Pricing in Ride-Sharing." *Expert Systems With Applications* 248: 123377. https://doi.org/10.1016/j.eswa.2024.124664.

Siddiq, A., and T. A. Taylor. 2022. "Ride-Hailing Platforms: Competition and Autonomous Vehicles." *Manufacturing & Service Operations Management* 24, no. 3: 1511–1528. https://doi.org/10.1287/msom.2021.1013.

Stogios, C., D. Kasraian, M. J. Roorda, and M. Hatzopoulou. 2019. "Simulating Impacts of Automated Driving Behavior and Traffic Conditions on Vehicle Emissions." *Transportation Research Part D: Transport and Environment* 76: 176–192. https://doi.org/10.1016/j.trd.2019.09.020.

Subramanian, R., S. Gupta, and B. Talbot. 2007. "Compliance Strategies Under Permits for Emissions." *Production and Operations Management* 16, no. 6: 763–779. https://doi.org/10.1111/j.1937-5956.2007.tb00294.x.

Sun, Z., and J. Liu. 2023. "Impacts of Differentiated Services and Competition on the Pricing Strategies of Ride-Hailing Platforms." *Managerial and Decision Economics* 44, no. 6: 3604–3624. https://doi.org/10.1002/mde.3898.

Taylor, T. A. 2018. "On-Demand Service Platforms." *Manufacturing & Service Operations Management* 20, no. 4: 704–720. https://doi.org/10.1287/msom.2017.0678.

Tripathy, M., J. Bai, and H. S. S. Heese. 2023. "Driver Collusion in Ride-Hailing Platforms." *Decision Sciences* 54, no. 4: 434–446. https://doi.org/10.1111/deci.12561.

Wang, H., Y. Li, Y. Liu, X. Hu, and J. Wang. 2025b. "Optimizing On-Demand Ride-Hailing Services in Two-Sided Coupled Markets With Impatient Riders." *Transportation Research Part E: Logistics and Transportation Review* 198: 104106. https://doi.org/10.1016/j.tre.2025.104106.

Wang, P., H. Jiang, Y. Cui, M. Zhao, Y. Ren, and L. Xu. 2025a. "Analysis of the Inter-City Shared Mobility System Based on Autonomous Electric Vehicles." *Renewable Energy* 239: 122025. https://doi.org/10.1016/j.renene.2024.122025.

Wang, X., X. Lin, and M. Li. 2021. "Aggregate Modeling and Equilibrium Analysis of the Crowdsourcing Market for Autonomous Vehicles." *Transportation Research Part C: Emerging Technologies* 132: 103362. https://doi.org/10.1016/j.trc.2021.103362.

Wang, X., and R. Zhang. 2022. "Carpool Services for Ride-Sharing Platforms: Price and Welfare Implications." *Naval Research Logistics* 69, no. 4: 550–565. https://doi.org/10.1002/nav.22030.

Wu, L., and G. F. N. Mvondo. 2025. "From Sci-Fi to Reality: Exploring User Satisfaction and Loyalty Toward Autonomous Vehicle Services Through an Extended Expectation-Confirmation Model." *Transportation Research Part F: Traffic Psychology and Behaviour* 113: 409–425. https://doi.org/10.1016/j.trf.2025.05.003.

Xie, N., Y. Chen, W. Tang, and X. M. Chen. 2025. "Multi-Agent Reinforcement Learning With Causal Communication for Ride-Sourcing Pricing in Mixed Autonomy Mobility." *Transportation Research Part C: Emerging Technologies* 176: 105164. https://doi.org/10.1016/j.trc.2025.105164.

Yang, F., L. Geng, and F. Shan. 2023. "Comparison of Three Participation Modes on Ride-Hailing Platforms." *Computers & Industrial Engineering* 183: 109473. https://doi.org/10.1016/j.cie.2023.109473.

Yang, L., T.-M. Choi, and X. Shi. 2025. "Risks in On-Demand Service Platform Operations: The Innovative Framework With SET Strategies." *IEEE Transactions on Engineering Management* 72: 2322–2329. https://doi.org/10.1109/TEM.2025.3567473.

Yao, F., J. Zhu, J. Yu, C. Chen, and X. M. Chen. 2020. "Hybrid Operations of Human Driving Vehicles and Automated Vehicles With Data-Driven Agent-Based Simulation." *Transportation Research Part D: Transport and Environment* 86: 102469. https://doi.org/10.1016/j.trd.2020.102469.

Yao, X., J. Liu, X. Wang, Y. Xiao, and G. Qi. 2025. "Transition Toward Driverless Robotaxi: Role of Social Anxiety, Perceived Safety, and Travel Habit." *Transportation Research Part F: Traffic Psychology and Behaviour* 109: 1402–1418. https://doi.org/10.1016/j.trf.2025.01.03.

Yi, Z., and J. Smart. 2021. "A Framework for Integrated Dispatching and Charging Management of an Autonomous Electric Vehicle Ride-Hailing Fleet." *Transportation Research Part D: Transport and Environment* 95: 102822. https://doi.org/10.1016/j.trd.2021.102822.

Yu, G., A. Liu, J. Zhang, and H. Sun. 2021. "Optimal Operations Planning of Electric Autonomous Vehicles via Asynchronous Learning in Ride-Hailing Systems." *Omega* 103: 102448. https://doi.org/10.1016/j.omega.2021.102448.

Yu, J. J., C. S. Tang, Z.-J. Max Shen, and X. M. Chen. 2020. "A Balancing Act of Regulating On-Demand Ride Services." *Management Science* 66, no. 7: 2975–2992. https://doi.org/10.1287/mnsc.2019.3351.

Zhang, C., J. Chen, and S. Raghunathan. 2022. "Two-Sided Platform Competition in a Sharing Economy." *Management Science* 68, no. 12: 8909–8932. https://doi.org/10.1287/mnsc.2022.4302.

Zhang, J.-H., R. Zhu, and J.-P. Wang. 2025. "Joint Re-Dispatching and Pricing on a Business-to-Consumer Ride-Hailing Platform." *Transportation Research Part E: Logistics and Transportation Review* 196: 103998. https://doi.org/10.1016/j.tre.2025.103998.

Zhang, X., H. Sun, X. Pei, L. Guan, and Z. Wang. 2024. "Evolution of Technology Investment and Development of Robotaxi Services."

Transportation Research Part E: Logistics and Transportation Review 188: 103615. https://doi.org/10.1016/j.tre.2024.103615.

Zhao, M., B. Li, J. Ren, and Z. Hao. 2023. "Competition Equilibrium of Ride-Sourcing Platforms and Optimal Government Subsidies Considering Customers' Green Preference Under Peak Carbon Dioxide Emissions." *International Journal of Production Economics* 255: 108679. https://doi.org/10.1016/j.ijpe.2022.108679.

Zhong, Y., T. Yang, B. Cao, and T. C. E. Cheng. 2022. "On-Demand Ride-Hailing Platforms in Competition With the Taxi Industry: Pricing Strategies and Government Supervision." *International Journal of Production Economics* 243: 108301. https://doi.org/10.1016/j.ijpe.2021.108301.

Zhou, Y., H. Yang, J. Ke, H. Wang, and X. Li. 2022. "Competition and Third-Party Platform-Integration in Ride-Sourcing Markets." *Transportation Research Part B: Methodological* 159: 76–103. https://doi.org/10.1016/j.trb.2021.08.002.

Supporting Information

Additional supporting information can be found online in the Supporting Information section. **Data S1:** Supplementary information.

Appendix A

Summary of Thresholds

TABLE A1 | Summary of thresholds used.

$$\tau_1 = \frac{(1-r)\alpha\lambda\phi\left((2+\alpha f-rt\alpha\beta(1+2\alpha f))+\sqrt{4+\alpha\left(\begin{array}{c} f(4-7\alpha f)-2rt\beta(2+\alpha f(5-6\alpha f)) \\ +r^2t^2\alpha\beta^2(1+4\alpha f(1-\alpha f)) \end{array}\right)}\right)}{16f(1-rt\alpha\beta)^2}$$

$$\tau_2 = \frac{(1-r)(1-rt\alpha\beta)\alpha\lambda\phi\left((4+2\alpha f-rt\alpha\beta(5+\alpha f))+\sqrt{16-\alpha\left(\begin{array}{c} rt\beta(40-17rt\alpha\beta)+7\alpha f^2(2-rt\alpha\beta)^2 \\ -2f(2-rt\alpha\beta)(4+3rt\alpha\beta) \end{array}\right)}\right)}{16(1-rt\alpha\beta)^2(2f-rt\beta(1+\alpha f))}$$

$$\tau_3 = \frac{(1-r)\alpha\lambda\phi\left((1+6\alpha f-rt\alpha\beta(4+3\alpha f))+\sqrt{1+\alpha(2-rt\alpha\beta)\left(6f-4rt\beta-3\alpha f^2(2-rt\alpha\beta)\right)}\right)}{16(1-rt\alpha\beta)(2f-rt\beta(1+\alpha f))}$$

$$\tau_4 = \frac{\alpha\lambda\phi\left((1+2t\alpha\beta)+\sqrt{(1+4t\alpha\beta(1-t\alpha\beta))}\right)}{16t\beta}$$

$$\tau_5 = \frac{\left((1-\phi-r(\beta-\phi))\left(4(1-rt\alpha\beta)^2-t\beta\alpha^3\lambda\phi(1-r)^2\right)-(\beta-\phi)(1-r)(1-rt\alpha\beta)\right)\alpha^2\lambda\phi}{(1-rt\alpha\beta)\left((1-\phi-r(\beta-\phi))\left(8t\alpha\beta\left(4(1-rt\alpha\beta)-\alpha^2\lambda\phi(1-r)\right)-\alpha^2\lambda\phi(1-rt\alpha\beta)\right)-8(\beta-\phi)(1-rt\alpha\beta)\right)}$$

$$\tau_6 = \frac{\alpha^2\lambda(1-r\beta)}{4(1-rt\alpha\beta)}$$

$$\tau_7 = \frac{\alpha^2\lambda\left((5-r-4r\beta)+\sqrt{\alpha(17+r(6-40\beta-r(7-8\beta(1+2\beta))))}\right)}{32(1-rt\alpha\beta)}$$

$$\tau_8 = \frac{\alpha^2\delta\lambda\phi(1-r)}{4(r(\alpha f(2-t\alpha\beta)-1)+(1-r)(2\alpha f-1)\delta)}$$

$$\tau_9 = \frac{\alpha\delta\lambda\phi(1-r)(\delta(1-r)+r(1-t\alpha\beta))\left(\begin{array}{c} (-2\alpha(r(t\beta-f)-\delta f(1-r))+(2-\alpha f)(r(1-t\alpha\beta)+\delta(1-r)))+ \\ \sqrt{\begin{array}{c} r^2\left(4+\alpha\left(\begin{array}{c} f(4-7\alpha f)+t^2\alpha\beta^2(8-\alpha f(8-\alpha f)) \\ -2t\beta(8-\alpha f(6+\alpha f)) \end{array}\right)\right)+ \\ 2(1-r)(4+\alpha f(4-7\alpha f)-t\alpha\beta(8-\alpha f(6+\alpha f)))r\delta \\ +(1-r)^2(4+\alpha f(4-7\alpha f))\delta^2 \end{array}} \end{array}\right)}{-16(r(t\beta-f)-\delta f(1-r))(\delta(1-r)+r(1-t\alpha\beta))^2}$$

$$\tau_{10} = \frac{\alpha^3\delta\lambda\phi(1-r)(2\delta f(1-r)-r(t\beta-f(2-t\alpha\beta)))}{8(r(1-\alpha f(2-t\alpha\beta))+\delta(1-r)(1-2\alpha f))(\delta(1-r)+r(1-t\alpha\beta))}$$

$$\tau_{11} = \frac{\alpha^2\lambda\phi\left(\begin{array}{c} \left(r\left(2(1-\beta)+2t\alpha\beta^2+\beta(3\delta-t\alpha(2+\delta(1-\phi)))-\delta(2+\phi)\right)+\delta(2+\phi-\beta(3-t\alpha(1-\phi)))\right)+ \\ \sqrt{\begin{array}{c} \delta^2\left(4+\beta^2(1+t\alpha(10-7t\alpha(1-\phi))(1-\phi))+(4-7\phi)\phi-2\beta(6-5\phi-t\alpha(1-\phi)(2-7\phi))\right)+ \\ 2r\delta\left(\begin{array}{c} 4(1-\delta)-2t\alpha\beta^3(3-t\alpha(1-\phi))+\phi(2-\delta(4-7\phi)) \\ +\beta^2(6-\delta+t\alpha(8-10\delta-t\alpha(2-7\delta(1-\phi))(1-\phi)+2\phi(2+5\delta))) \\ -2\beta(5+\phi-\delta(6-5\phi)+t\alpha(1+2\phi+\delta(1-\phi)(2-7\phi))) \end{array}\right)+ \\ +r^2\left(\begin{array}{c} 4\left(1+t^2\alpha^2\beta^4+t\alpha\beta^3(3\delta-2-t\alpha(2+\delta(1-\phi)))-\delta(2+\phi)\right)+\delta^2(4+(4-7\phi)\phi) \\ +\beta^2\left(4-(12-\delta)\delta+t^2\alpha^2(4+\delta(4-7\delta(1-\phi))(1-\phi))+2t\alpha(8-\delta(8+4\phi-5\delta(1-\phi)))\right) \\ +2\beta(\delta(10-6\delta+2\phi+5\delta\phi)-4+t\alpha(\delta(2+4\phi+\delta(1-\phi)(2-7\phi))-4)) \end{array}\right) \end{array}} \end{array}\right)}{16(\delta(1-r)+r(1-t\alpha\beta))(\beta(t\alpha-1)+\phi(1-t\alpha\beta))}$$

$$\tau_{12} = \frac{\alpha^2\lambda\phi\left(\begin{array}{c} ((1-\beta)(1-rt\alpha\beta)+(t\alpha-1)(1-r)\beta+(1-t\alpha\beta)(1-r)\phi) \\ +\sqrt{(1-\beta)(1-t\alpha\beta)(1-rt\alpha\beta)(1-\phi-r(\beta-\phi))} \end{array}\right)}{8(1-rt\alpha\beta)(\beta(t\alpha-1)+\phi(1-t\alpha\beta))}$$

(Continues)

TABLE A1 | (Continued)

$$\tau_{13} = \frac{\alpha^2\lambda\phi\left(\begin{array}{c}(1-\beta)(\delta(1-r)+r(1-t\alpha\beta))+\delta(1-r)(\beta(t\alpha-1)+\phi(1-t\alpha\beta))\\+\sqrt{(1-\beta)(1-t\alpha\beta)(\delta(1-r)+r(1-t\alpha\beta))(\delta(1-\phi)(1-r)+r(1-\beta))}\end{array}\right)}{8(\delta(1-r)+r(1-t\alpha\beta))(\beta(t\alpha-1)+\phi(1-t\alpha\beta))}$$

$$\tau_{14} = \frac{\alpha\left(\begin{array}{c}\left(\begin{array}{c}\alpha\lambda\phi(1-r)(1-\alpha f)((1-\beta)(1-rt\alpha\beta)+(t\alpha-1)(1-r)\beta+(1-t\alpha\beta)(1-r)\phi)\\-2c(1-t\alpha\beta)(1-rt\alpha\beta)(1-\phi-r(\beta-\phi))\end{array}\right)\\+(\alpha\lambda\phi(1-r)(1-\alpha f)-2c(1-rt\alpha\beta))\sqrt{(1-\beta)(1-t\alpha\beta)(1-rt\alpha\beta)(1-\phi-r(\beta-\phi))}\end{array}\right)}{8(1-r)(1-rt\alpha\beta)(1-\alpha f)(\beta(t\alpha-1)+\phi(1-t\alpha\beta))}$$

$$\tau_{15} = \frac{\alpha\left(\begin{array}{c}\left(\begin{array}{c}\alpha\delta\lambda\phi(1-r)(1-\alpha f)((1-\beta)(\delta(1-r)+r(1-t\alpha\beta))+\delta(1-r)(\beta(t\alpha-1)+\phi(1-t\alpha\beta)))\\-2c(1-t\alpha\beta)(\delta(1-r)+r(1-t\alpha\beta))(\delta(1-\phi)(1-r)+r(1-\beta))\end{array}\right)\\+\left(\begin{array}{c}(\alpha\delta\lambda\phi(1-r)(1-\alpha f)-2c(\delta(1-r)+r(1-t\alpha\beta)))\\\sqrt{(1-\beta)(1-t\alpha\beta)(\delta(1-r)+r(1-t\alpha\beta))(\delta(1-\phi)(1-r)+r(1-\beta))}\end{array}\right)\end{array}\right)}{8\delta(1-r)(1-\alpha f)(\delta(1-r)+r(1-t\alpha\beta))(\beta(t\alpha-1)+\phi(1-t\alpha\beta))}$$

τ_{16} is the solution of $\frac{\partial\pi_p^{d*}}{\partial\delta} = 0$ in Section 6.2

$$\phi_1 = \frac{2(1-rt\alpha\beta)\tau-2\sqrt{\tau(1-rt\alpha\beta)\left(2\alpha^2\lambda(1-r\beta)-7\tau(1-rt\alpha\beta)\right)}}{\alpha^2\lambda(1-r)}$$

$$\mu^s = \frac{-2(2-rt\alpha\beta)\tau+(1-r)\alpha^2\lambda\phi+2\sqrt{\tau\left((2-rt\alpha\beta)^2\tau-\alpha^2\lambda\phi(1-r)(1-rt\alpha\beta)\right)}}{(1-r)\alpha^2\lambda\phi}$$

$$\mu^d = \frac{\left(\begin{array}{c}-2(2\delta(1-r)+r(2-t\alpha\beta))\tau+(1-r)\alpha^2\delta\lambda\phi\\+2\sqrt{\tau\left((2\delta(1-r)+r(2-t\alpha\beta))^2\tau-\alpha^2\delta\lambda\phi(1-r)(\delta(1-r)+r(1-t\alpha\beta))\right)}\end{array}\right)}{(1-r)\alpha^2\delta\lambda\phi}$$

$$\overline{\delta} = \frac{r(t\beta-f(4-3t\alpha\beta))\left(8\tau-\alpha^2\lambda\phi\right)+r\sqrt{16t^2\beta^2(1-\alpha f)^2\tau\left(4\tau-\alpha^2\lambda\phi\right)+\alpha^4\lambda^2\phi^2(t\beta-f(4-3t\alpha\beta))^2}}{8f(1-r)\left(4\tau-\alpha^2\lambda\phi\right)}$$